

MODELAGEM INFORMACIONAL

Modelo Dimensional



- Modelo dimensional**
- Data Marts**

INTRODUCTION

- *ER modeling*
 - *A predominant technique for visualizing database requirements, used extensively for conceptual modeling of operational databases*
- *Relational modeling*
 - *Standard method for logical modeling of operational databases*
- *Both of these techniques can also be used during the development of data warehouses and data marts*
- *Dimensional modeling*
 - *A modeling technique tailored specifically for analytical database design purposes*
 - *Regularly used in practice for modeling data warehouses and data marts*

DIMENSIONAL MODELING

■ *Dimensional modeling*

- *A data design methodology used for designing subject-oriented analytical databases, such as data warehouses or data marts*
- *Commonly, dimensional modeling is employed as a relational data modeling technique*
- *In addition to using regular relational concepts (primary keys, foreign keys, integrity constraints, etc.) dimensional modeling distinguishes two types of tables:*
 - **Dimensions**
 - **Facts**

DIMENSIONAL MODELING

- ***Dimension tables (dimensions)***
 - *Contain descriptions of the business, organization, or enterprise to which the subject of analysis belongs*
 - *Columns in dimension tables contain descriptive information that is often textual (e.g., product brand, product color, customer gender, customer education level), but can also be numeric (e.g., product weight, customer income level)*
 - *This information provides a basis for analysis of the subject*

DIMENSIONAL MODELING

■ *Fact tables*

- *Contain measures related to the subject of analysis and the foreign keys (associating fact tables with dimension tables)*
- *The measures in the fact tables are typically numeric and are intended for mathematical computation and quantitative analysis*

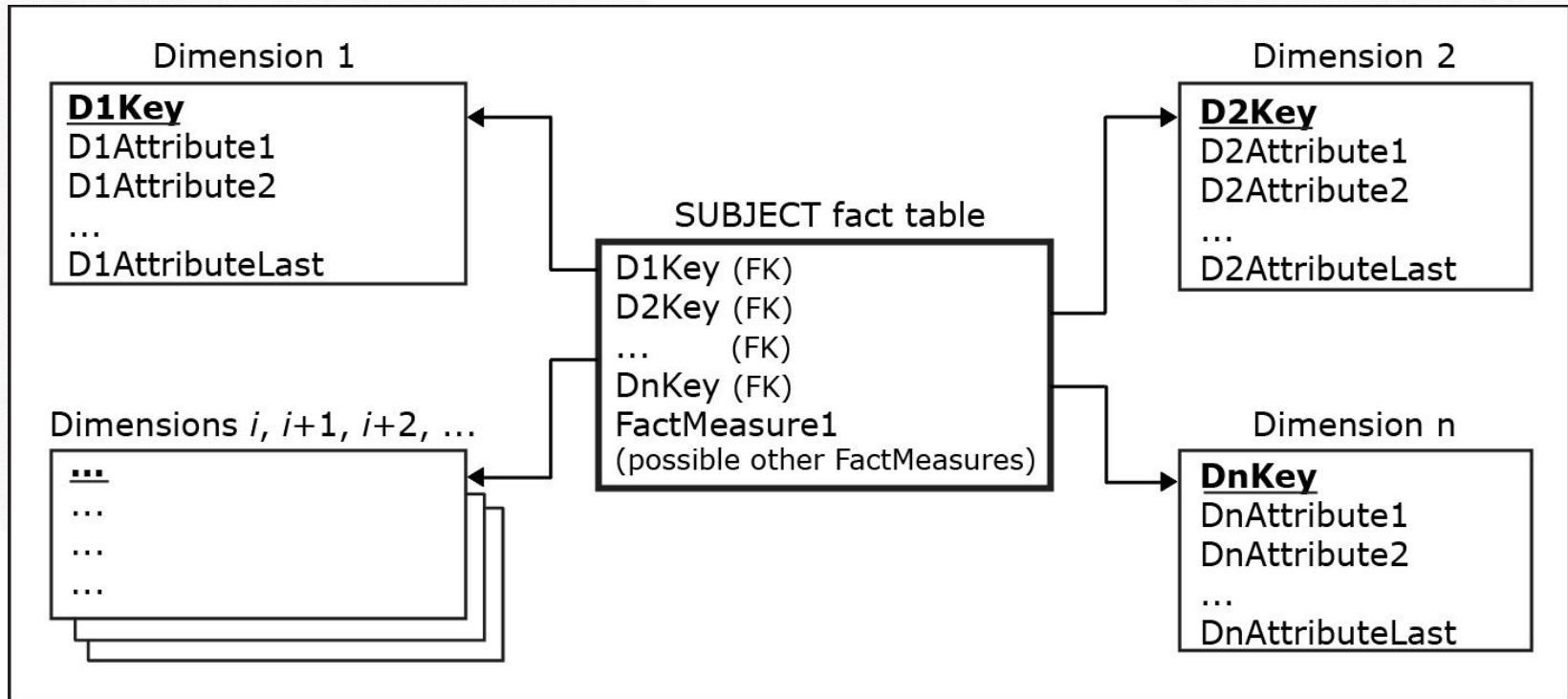
DIMENSIONAL MODELING

- **Star schema**

- *The result of dimensional modeling is a dimensional schema containing facts and dimensions*
- *The dimensional schema is often referred to as the star schema*

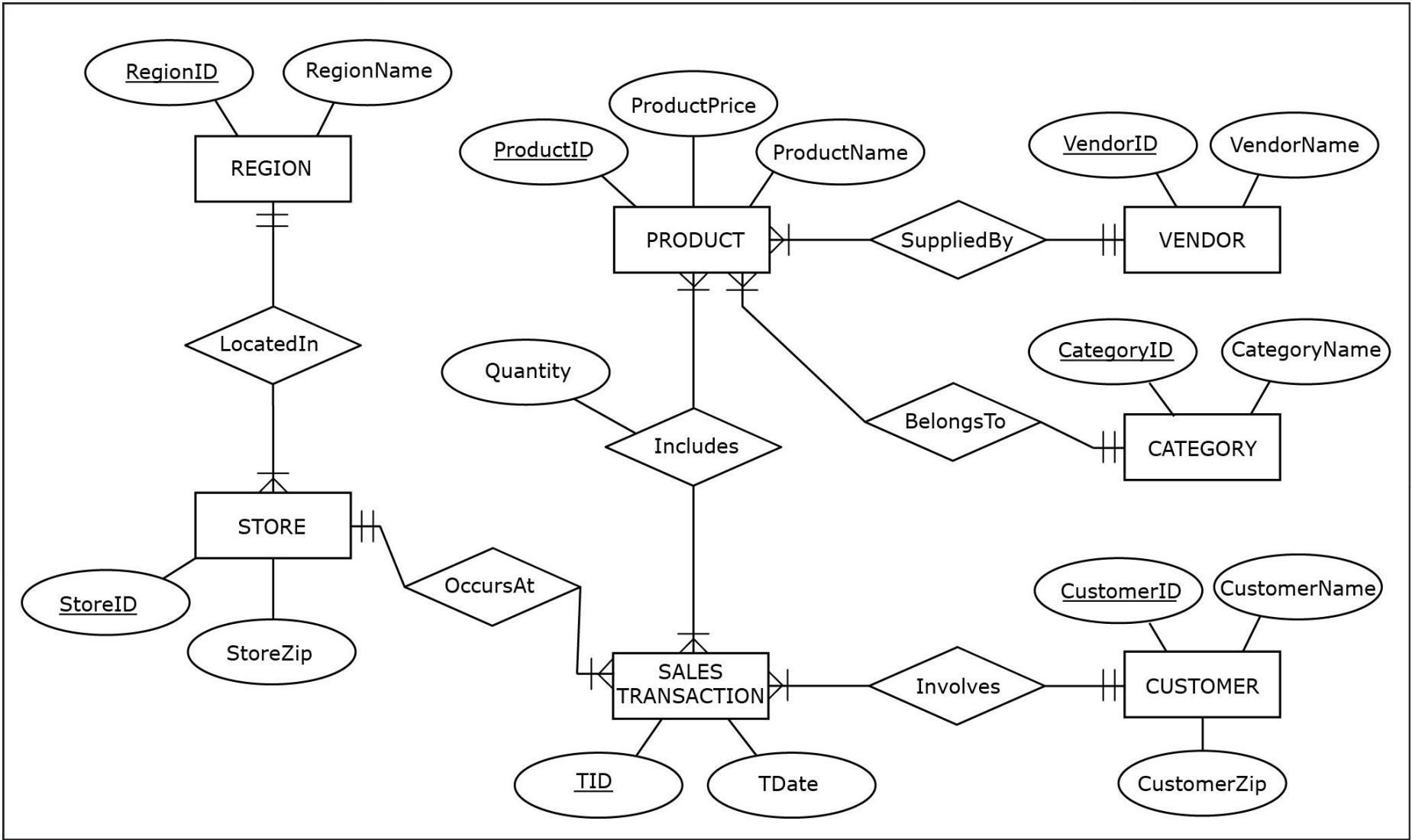
DIMENSIONAL MODELING

A dimensional model (star schema)



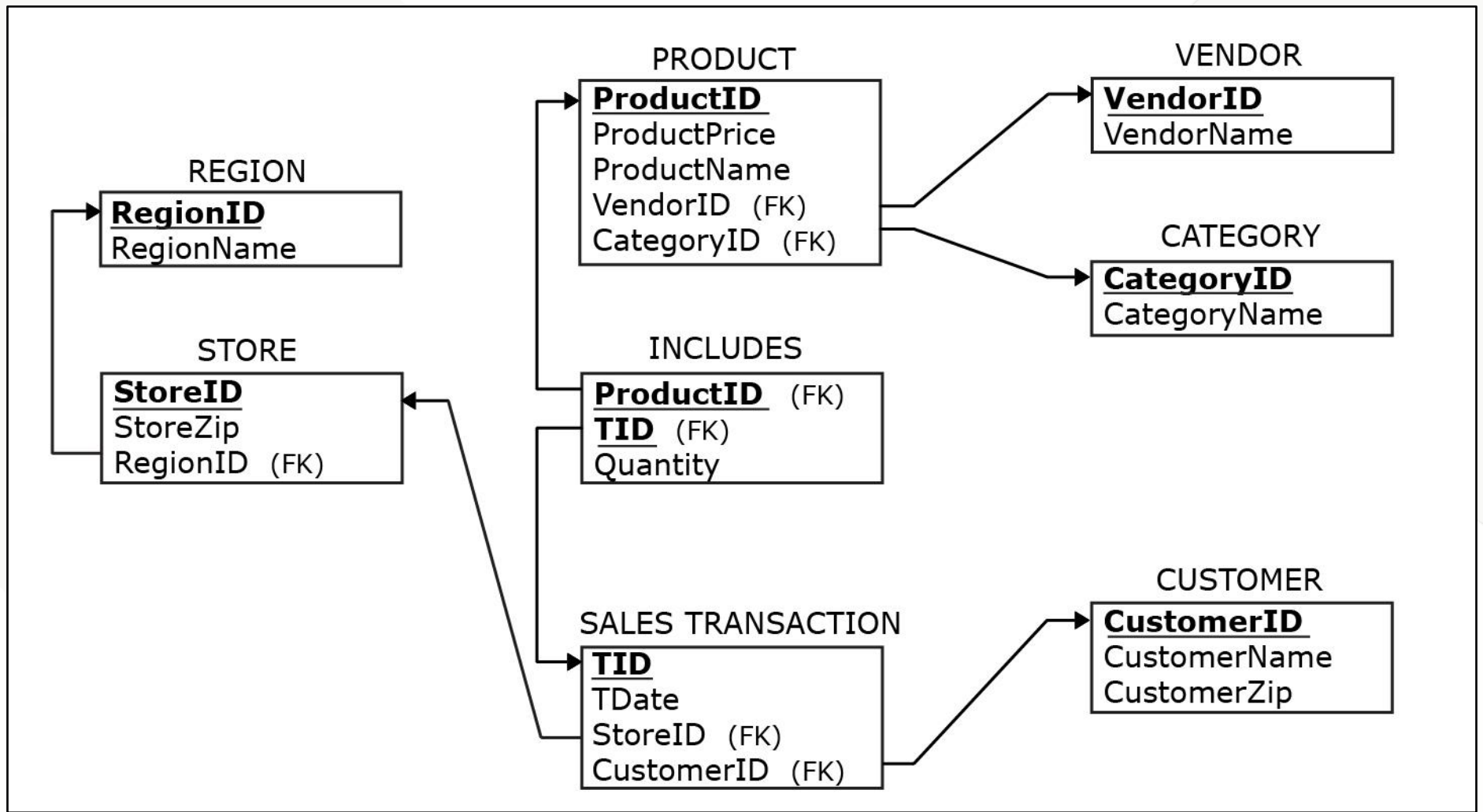
Initial Example: Dimensional Model Based on A Single Source

ER diagram : ZAGI Retail Company Sales Department Database (Source)



Initial Example: Dimensional Model Based on A Single Source

Relational schema : ZAGI Retail Company Sales Department Database (Source)



Initial Example: Dimensional Model Based on A Single Source

Data records: ZAGI Retail Company Sales Department Database (Source)

REGION

RegionID	RegionName
C	Chicagoland
T	Tristate

INCLUDES

ProductID	TID	Quantity
1X1	T111	1
2X2	T222	1
3X3	T333	5
1X1	T333	1
4X4	T444	1
2X2	T444	2
4X4	T555	4
5X5	T555	2
6X6	T555	1

VENDOR

VendorID	VendorName
PG	Pacific Gear
MK	Mountain King

STORE

StoreID	StoreZip	RegionID
S1	60600	C
S2	60605	C
S3	35400	T

CATEGORY

CategoryID	CategoryName
CP	Camping
FW	Footwear

SALES TRANSACTION

TID	CustomerID	StoreID	TDate
T111	1-2-333	S1	1-Jan-2020
T222	2-3-444	S2	1-Jan-2020
T333	1-2-333	S3	2-Jan-2020
T444	4-5-666	S3	2-Jan-2020
T555	2-3-444	S3	2-Jan-2020

CUSTOMER

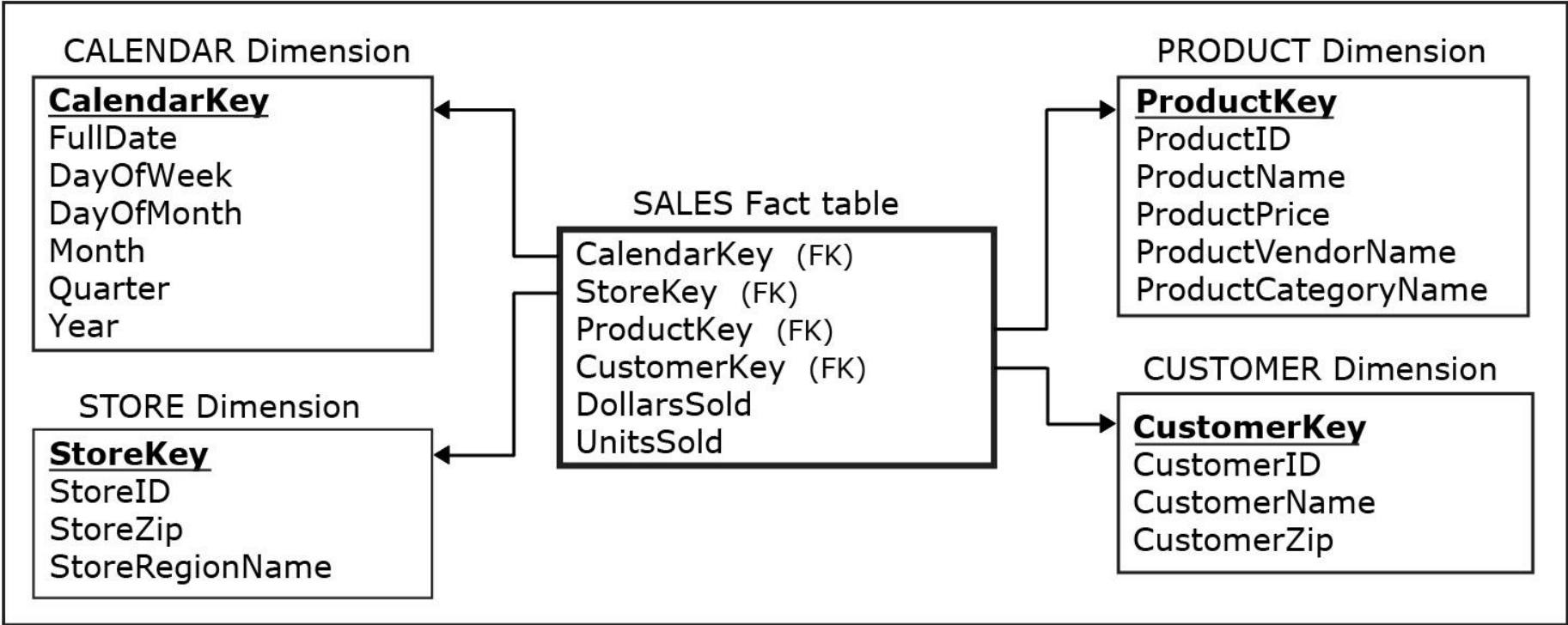
CustomerID	CustomerName	CustomerZip
1-2-333	Tina	60137
2-3-444	Tony	60611
3-4-555	Pam	35401

PRODUCT

ProductID	ProductName	ProductPrice	VendorID	CategoryID
1X1	Zzz Bag	\$100	PG	CP
2X2	Easy Boot	\$70	MK	FW
3X3	Cosy Sock	\$15	MK	FW
4X4	Dura Boot	\$90	PG	FW
5X5	Tiny Tent	\$150	MK	CP
6X6	Biggy Tent	\$250	MK	CP

Initial Example: Dimensional Model Based on A Single Source

*ZAGI Retail Company dimensional model for the subject **sales***



DIMENSIONAL MODELING

■ *Star schema*

- *In the star schema, the chosen subject of analysis is represented by a fact table*
- *Designing the star schema involves considering which dimensions to use with the fact table representing the chosen subject*
- *For every dimension under consideration, two questions must be answered:*
 - ***Question 1: Can the dimension table be created based on the existing data sources?***
 - ***Question 2: Can the dimension table be useful for the analysis of the chosen subject?***

Initial Example: Dimensional Model Based on A Single Source

*ZAGI Retail Company dimensional model for the subject **sales**, populated with the data from the operational data source*

CALENDAR Dimension

<u>Calendar Key</u>	FullDate	DayOfWeek	DayOf Month	Month	Qtr	Year
1	1-Jan-2020	Wednesday	1	January	Q1	2020
2	2-Jan-2020	Thursday	2	January	Q1	2020

STORE Dimension

<u>Store Key</u>	Store ID	StoreZip	StoreRegion Name
1	S1	60600	Chicagoland
2	S2	60605	Chicagoland
3	S3	35400	Tristate

CUSTOMER Dimension

<u>Customer Key</u>	Customer ID	Customer Name	Customer Zip
1	1-2-333	Tina	60137
2	2-3-444	Tony	60611
3	3-4-555	Pam	35401

PRODUCT Dimension

<u>Product Key</u>	Product ID	Product Name	Product Price	Product Vendor Name	Product Category Name
1	1X1	Zzz Bag	\$100	Pacifica Gear	Camping
2	2X2	Easy Boot	\$70	Mountain King	Footwear
3	3X3	Cosy Sock	\$15	Mountain King	Footwear
4	4X4	Dura Boot	\$90	Pacifica Gear	Footwear
5	5X5	Tiny Tent	\$150	Mountain King	Camping
6	6X6	Biggy Tent	\$250	Mountain King	Camping

SALES Fact table

<u>Calendar Key</u>	<u>Store Key</u>	<u>Product Key</u>	<u>Customer Key</u>	Dollars Sold	Units Sold
1	1	1	1	\$100	1
1	2	2	2	\$70	1
2	3	3	1	\$75	5
2	3	1	1	\$100	1
2	3	4	3	\$90	1
2	3	2	3	\$140	2
2	3	4	2	\$360	4
2	3	5	2	\$300	2
2	3	6	2	\$250	1

DIMENSIONAL MODELING

■ *Characteristics of dimensions and facts*

- *A typical dimension contains relatively static data, while in a typical fact table, records are added continually, and the table rapidly grows in size.*
- *In a typical dimensionally modeled analytical database, dimension tables have orders of magnitude fewer records than fact tables*

DIMENSIONAL MODELING

▪ *Surrogate key / Chave substituta*

- *Typically, in a star schema all dimension tables are given a simple, non-composite system-generated key, also called a surrogate key*
- *Values for the surrogate keys are typically simple auto-increment integer values*
- *Surrogate key values have no meaning or purpose except to give each dimension a new column that serves as a primary key within the dimensional model instead of the operational key*

Query A: *Compare the quantities of sold products on Saturdays in the category Camping provided by the vendor Pacifica Gear within the Tristate region between the 1st and 2nd quarter of the year 2020*

Example query - Query A, dimensional version

```
SELECT      SUM(SA.UnitsSold)
,            P.ProductCategoryName
,            P.ProductVendorName
,            C.DayofWeek
,            C.Qtr
FROM
            Calendar C
,            Store      S
,            Product    P
,            Sales      SA
WHERE

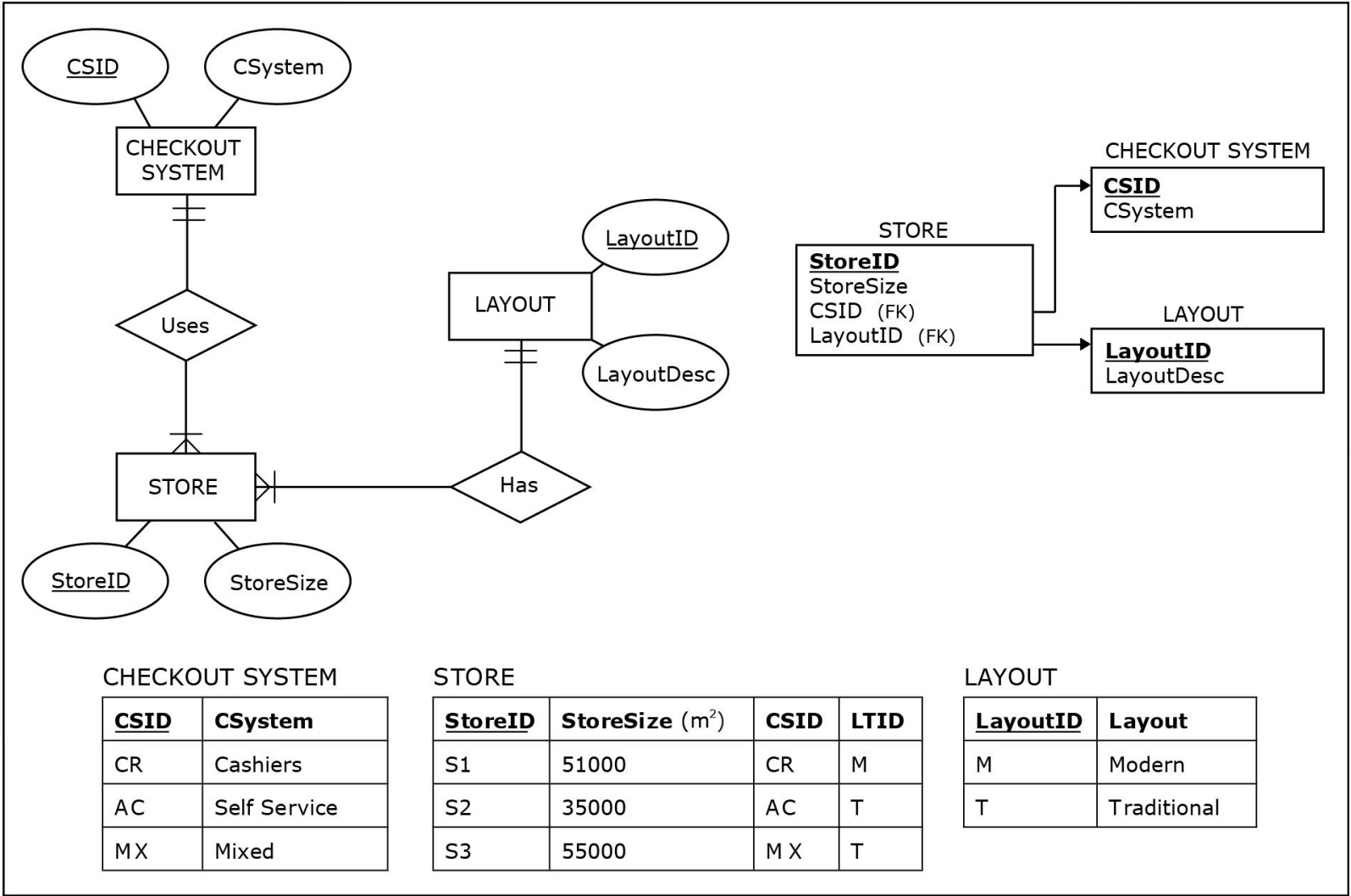
            C.CalendarKey = SA.CalendarKey
AND        S.StoreKey = SA.StoreKey
AND        P.ProductKey = SA.ProductKey
AND        P.ProductVendorName = 'Pacifica Gear'
AND        P.ProductCategoryName = 'Camping'
AND        S.StoreRegionName = 'Tristate'
AND        C.DayofWeek = 'Saturday'
AND        C.Year = 2020
AND        C.Qtr IN ( 'Q1', 'Q2' )
GROUP BY
            P.ProductCategoryName,
            P.ProductVendorName,
            C.DayofWeek,
            C.Qtr;
```

Example query - Query A, nondimensional version

```
SELECT      SUM( I.Quantity )
,            C.CategoryName
,            V.VendorName
,            EXTRACTWEEKDAY(ST.Date)
,            EXTRACTQUARTER(ST.Date)
FROM
            Region      R
,            Store      S
,            SalesTransaction ST
,            Includes   I
,            Product    P
,            Vendor     V
,            Category   C
WHERE
            R.RegionID = S.RegionID
AND        S.StoreID = ST.StoreID
AND        ST.Tid = SV.Tid
AND        I.ProductID = P.ProductID
AND        P.VendorID = V.VendorID
AND        P.CateoryID = C.CategoryID
AND        V.VendorName = 'Pacifica Gear'
AND        C.CategoryName = 'Camping'
AND        R.RegionName = 'Tristate'
AND        EXTRACTWEEKDAY(St.Date) = 'Saturday'
AND        EXTRACTYEAR(ST.Date) = 2020
AND        EXTRACTQUARTER(ST.Date) IN ( 'Q1', 'Q2' )
GROUP BY
            C.CategoryName,
            V.VendorName,
            EXTRACTWEEKDAY(ST.Date),
            EXTRACTQUARTER(ST.Date);
```


Expanded Example: Dimensional Model Based on Multiple Sources

ZAGI Retail Company Facilities Department Database (Source 2)



Expanded Example: Dimensional Model Based on Multiple Sources

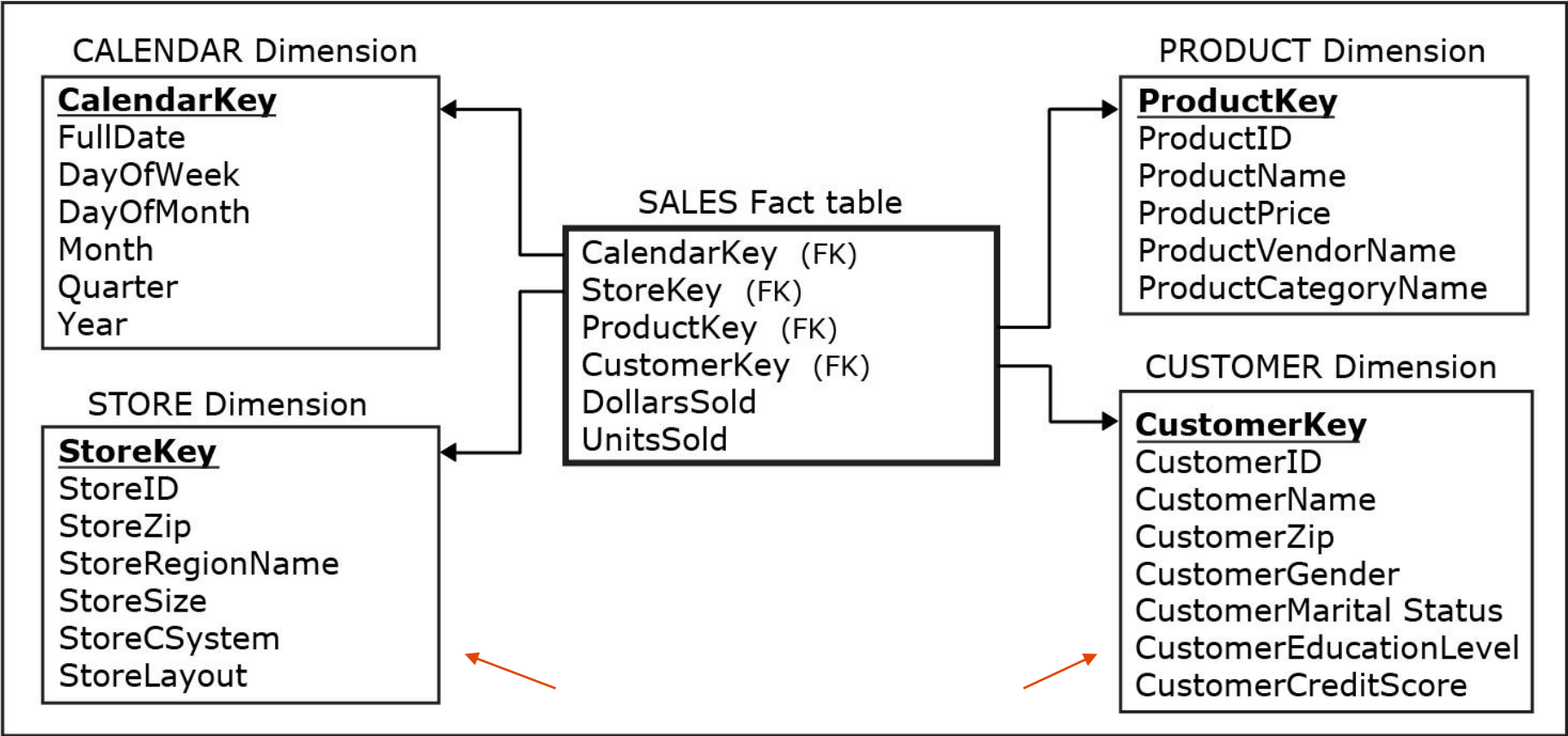
Customer Demographic Data Table - external source acquired from a market research company (Source 3)

EXTERNAL CUSTOMER TABLE					
<u>CustomerID</u>	Customer Name	Gender	Marital Status	Education Level	Credit Score
1-2-333	Tina	Female	Single	College	700
2-3-444	Tony	Male	Single	High School	650
3-4-555	Pam	Female	Married	College	623

Obs.: Exemplo ilustrativo, abordaremos LGPD no final da disciplina

Expanded Example: Dimensional Model Based on Multiple Sources

ZAGI Retail Company dimensional model for the subject *sales*



Expanded Example: Dimensional Model Based on Multiple Sources

ZAGI Retail Company dimensional model for the subject **sales** , populated with the data from the three sources

CALENDAR Dimension

<u>Calendar Key</u>	FullDate	DayOfWeek	DayOf Month	Month	Qtr	Year
1	1-Jan-2020	Wednesday	1	January	Q1	2020
2	2-Jan-2020	Thursday	2	January	Q1	2020

STORE Dimension

<u>Store Key</u>	StoreID	StoreZip	StoreRegion Name	StoreSize (m ²)	StoreCSystem	StoreLayout
1	S1	60600	Chicagoland	51000	Cashiers	Modern
2	S2	60605	Chicagoland	35000	Self Service	Traditional
3	S3	35400	Tristate	55000	Mixed	Traditional

SALES Fact table

<u>Calendar Key</u>	<u>Store Key</u>	<u>Product Key</u>	<u>Customer Key</u>	Dollars Sold	Units Sold
1	1	1	1	\$100	1
1	2	2	2	\$70	1
2	3	3	1	\$75	5
2	3	1	1	\$100	1
2	3	4	3	\$90	1
2	3	2	3	\$140	2
2	3	4	2	\$360	4
2	3	5	2	\$300	2
2	3	6	2	\$250	1

CUSTOMER Dimension

<u>Customer Key</u>	CustomerID	Customer Name	Customer Zip	Customer Marital Status	Customer Gender	Customer Education Level	Customer Credit Score
1	1-2-333	Tina	60137	Single	Female	College	700
2	2-3-444	Tony	60611	Single	Male	High School	650
3	3-4-555	Pam	35401	Married	Female	College	623

PRODUCT Dimension

<u>Product Key</u>	ProductID	ProductName	Product Price	Product VendorName	Product CategoryName
1	1X1	Zzz Bag	\$100	Pacifica Gear	Camping
2	2X2	Easy Boot	\$70	Mountain King	Footwear
3	3X3	Cosy Sock	\$15	Mountain King	Footwear
4	4X4	Dura Boot	\$90	Pacifica Gear	Footwear
5	5X5	Tiny Tent	\$150	Mountain King	Camping
6	6X6	Biggy Tent	\$250	Mountain King	Camping

Query B: *Compare the quantities of sold products to male customers in Modern stores on Saturdays in the category Camping provided by the vendor Pacifica Gear within the Tristate region between the 1st and 2nd quarter of the year 2020.*

Query B: *Compare as quantidades de produtos vendidos a clientes do sexo masculino nas lojas de lay-out moderno aos sábados, na categoria Camping fornecido pelo vendedor Pacifica Gear, na região de Tristate entre o 1º e o 2º trimestre do ano 2020.*

Example query - Query B, dimensional version

```
SELECT      SUM(SA.UnitsSold)
,            P.ProductCategoryName
,            P.ProductVendorName
,            C.DayofWeek
,            C.Qtr
FROM
            Calendar  C
,            Store    S
,            Product  P

,            Customer CU
,            Sales    SA
WHERE

            C.CalendarKey = SA.CalendarKey
AND        S.StoreKey = SA.StoreKey
AND        P.ProductKey = SA.ProductKey
AND        CU.CustomerKey = SA.CustomerKey
AND        P.ProductVendorName = 'Pacifica Gear'
AND        P.ProductCategoryName = 'Camping'
AND        S.StoreRegionName = 'Tristate'
AND        C.DayofWeek = 'Saturday'
AND        C.Year = 2020
AND        C.Qtr IN ( 'Q1', 'Q2' )
AND        S.StoreLayout = 'Modern'
AND        CU.Gender = 'Male'
GROUP BY
            P.ProductCategoryName,
            P.ProductVendorName,
            C.DayofWeek,
            C.Qtr;
```

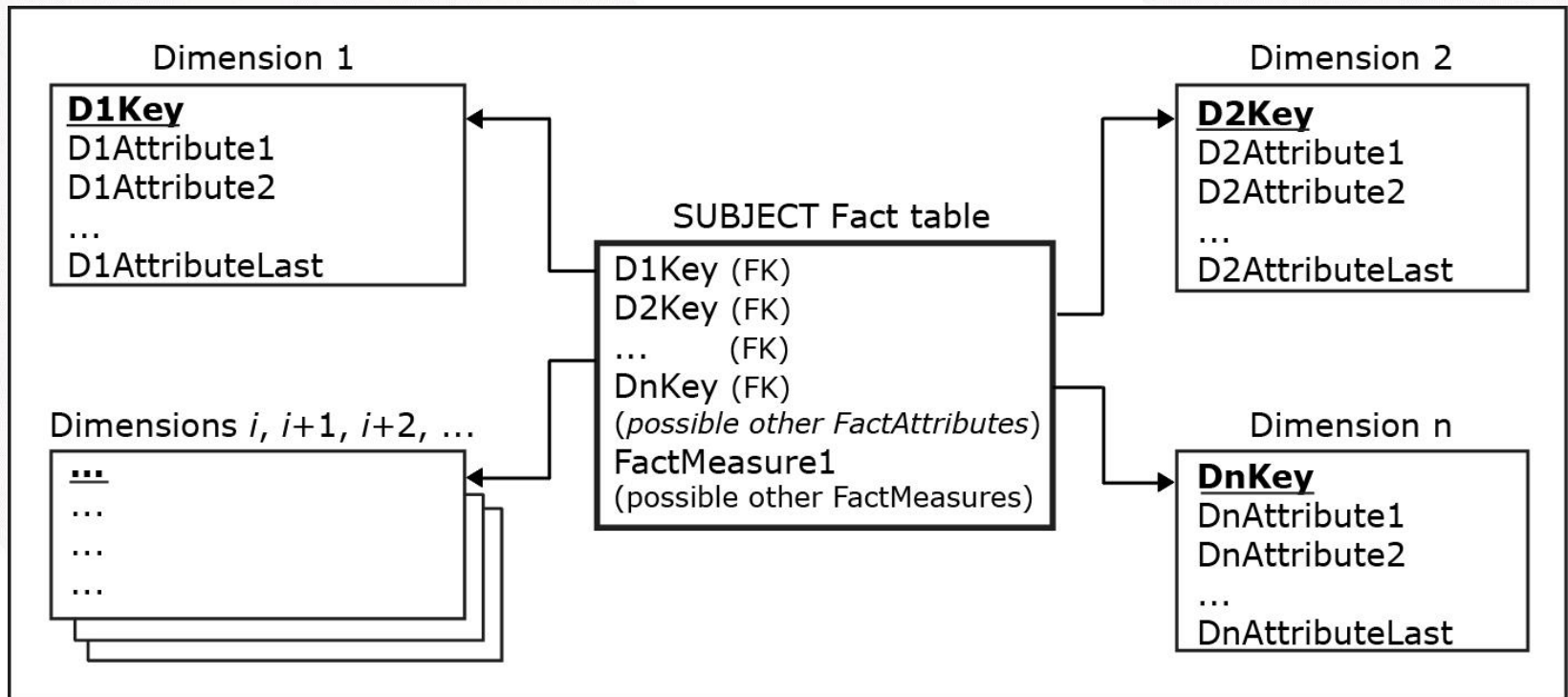
DIMENSIONAL MODELING

▪ *Additional possible fact attributes*

- *A fact table contains*
 - **Foreign keys connecting the fact table to the dimension tables**
 - **The measures related to the subject of analysis**
- *In addition to the measures related to the subject of analysis, in certain cases fact tables can contain other attributes that are not measures*
- *Two of the most typical additional attributes that can appear in the fact table are:*
 - **Transaction identifier**
 - **Transaction time**

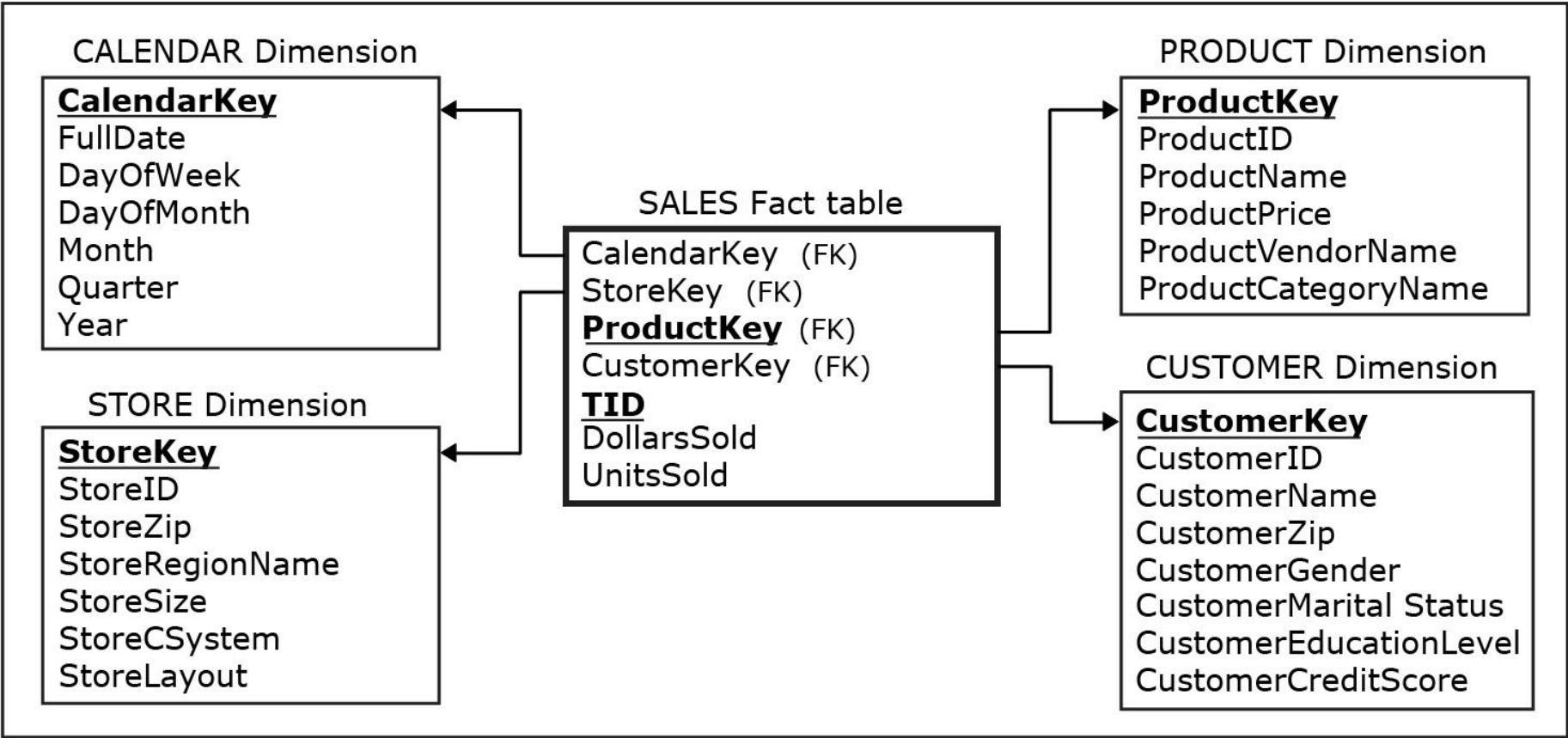
DIMENSIONAL MODELING

Additional possible fact attributes



Expanded Example: Dimensional Model with Transaction Identifier

ZAGI Retail Company dimensional model for the subject sales with *transaction identifier* included

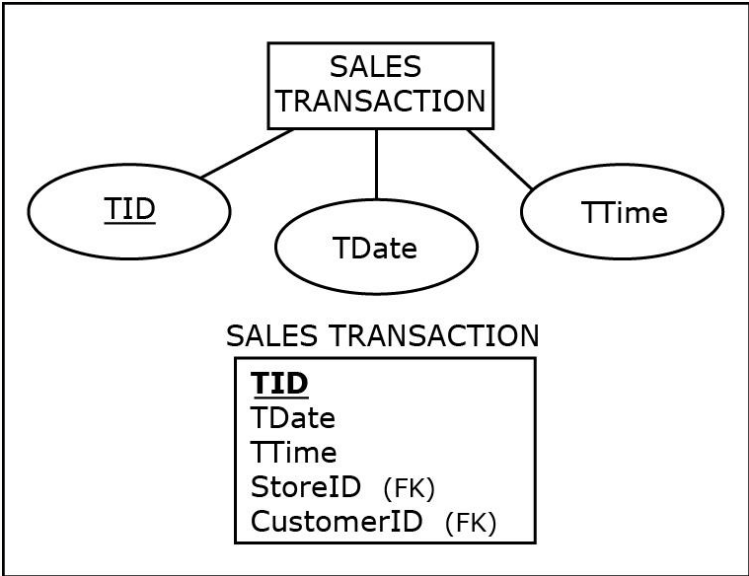


Fact table in the ZAGI Retail Company dimensional model for the subject sales, populated with the data, including the transaction identifier values

SALES Fact table						
Calendar Key	Store Key	Product Key	Customer Key	TID	Dollars Sold	Units Sold
1	1	1	1	T111	\$100	1
1	2	2	2	T222	\$70	1
2	3	3	1	T333	\$75	5
2	3	1	1	T333	\$100	1
2	3	4	3	T444	\$90	1
2	3	2	3	T444	\$140	2
2	3	4	2	T555	\$360	4
2	3	5	2	T555	\$300	2
2	3	6	2	T555	\$250	1

Expanded Example: Dimensional Model with Transaction Time

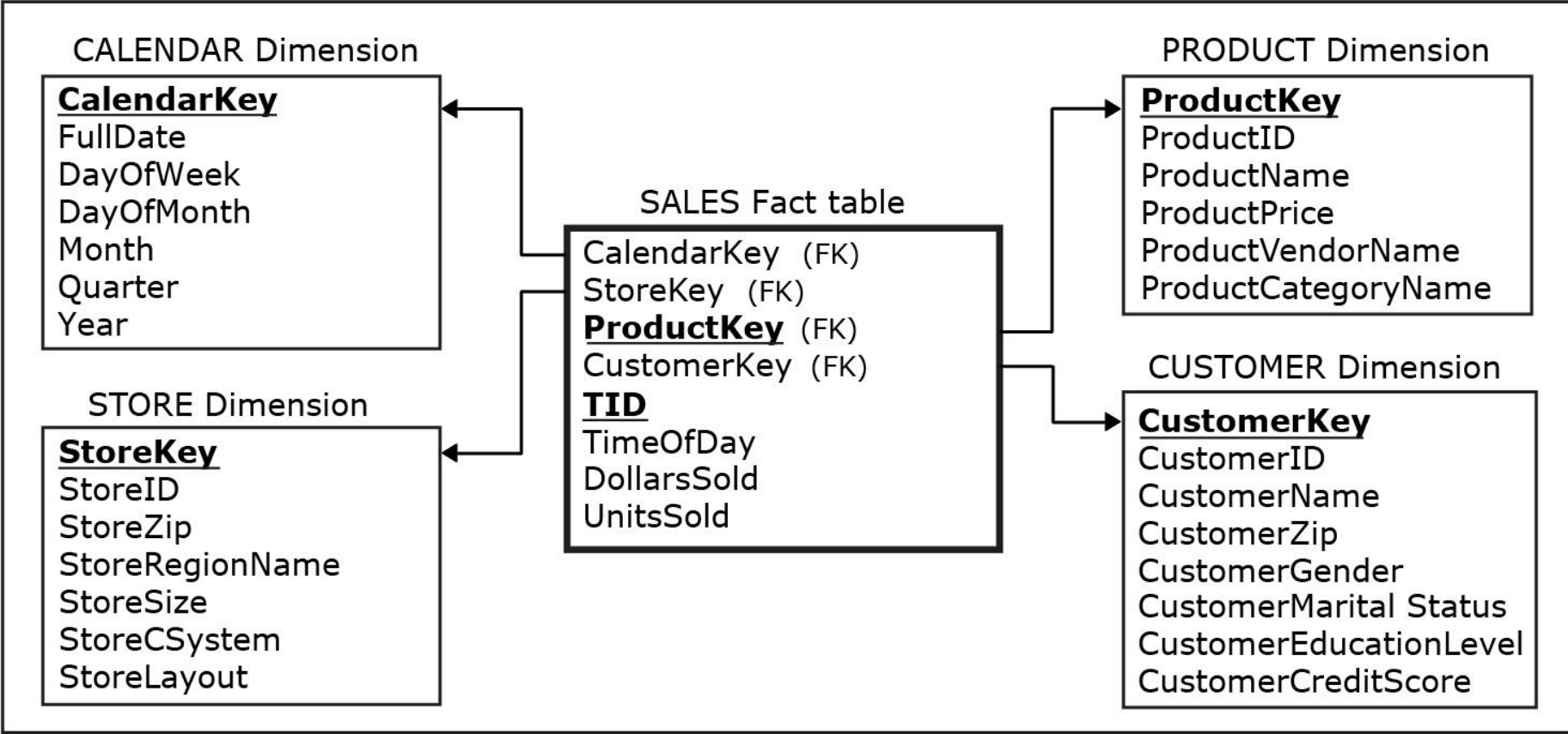
ER diagram : Transaction *time* attribute added to SALESTRANSACTION in the ZAGI Retail Company Sales Department Database (Source 1).



SALESTRANSACTION				
<u>TID</u>	CustomerID	StoreID	TDate	TTime
T111	1-2-333	S1	1-Jan-2020	8:23:59 AM
T222	2-3-444	S2	1-Jan-2020	8:24:30 AM
T333	1-2-333	S3	2-Jan-2020	8:15:08 AM
T444	4-5-666	S3	2-Jan-2020	8:20:33 AM
T555	2-3-444	S3	2-Jan-2020	8:30:00 AM

Expanded Example: Dimensional Model with Transaction Time

*ZAGI Retail Company dimensional model for the subject sales with **time** included*



Expanded Example: Dimensional Model Based on Multiple Sources

Fact table in the ZAGI Retail Company dimensional model for the subject sales, populated with the data, including the time values

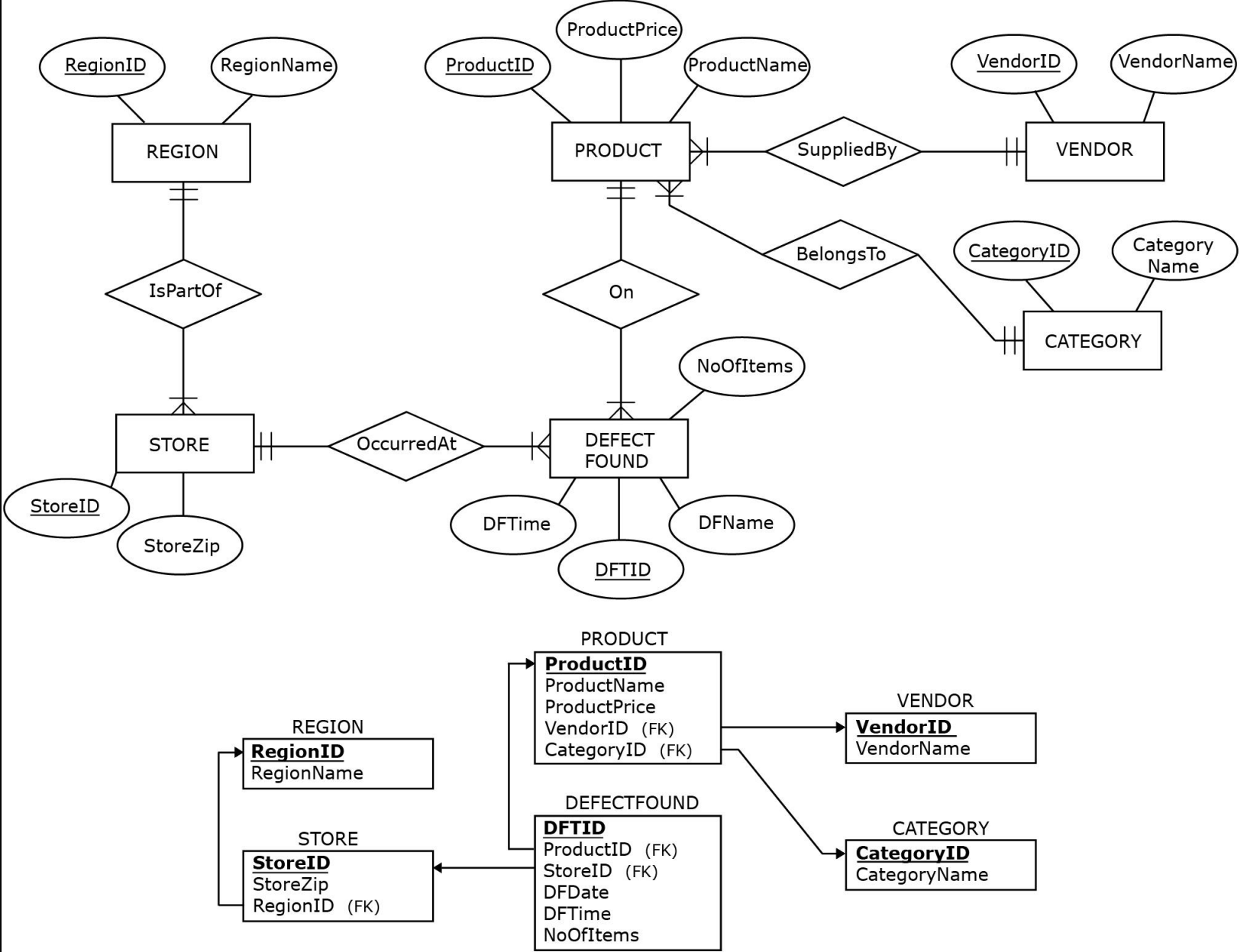
SALES Fact table							
CalendarKey	StoreKey	ProductKey	CustomerKey	TID	TimeOfDay	DollarsSold	UnitsSold
1	1	1	1	T111	8:23:59 AM	\$100	1
1	2	2	2	T222	8:24:30 AM	\$70	1
2	3	3	1	T333	8:15:08 AM	\$75	5
2	3	1	1	T333	8:15:08 AM	\$100	1
2	3	4	3	T444	8:20:33 AM	\$90	1
2	3	2	3	T444	8:20:33 AM	\$140	2
2	3	4	2	T555	8:30:00 AM	\$360	4
2	3	5	2	T555	8:30:00 AM	\$300	2
2	3	6	2	T555	8:30:00 AM	\$250	1

DIMENSIONAL MODELING

- **Multiple facts in a dimensional model**
 - *When multiple subjects of analysis can share the same dimensions, a dimensional model contains more than one fact table*
 - *A dimensional model with multiple fact tables is referred to as a **constellation** or **galaxy of stars***
 - *This approach enables:*
 - **Quicker development of analytical databases for multiple subjects of analysis, because dimensions are re-used instead of duplicated**
 - **Straightforward cross-fact analysis**

Expanded Example: Dimensional Model Based on Multiple Sources

ER diagram : ZAGI Retail Company Quality Control Database (Source 4)



Expanded Example: Dimensional Model Based on Multiple Sources

Data records: ZAGI Retail Company Quality Control Database (Source 4)

REGION

<u>RegionID</u>	RegionName
C	Chicagoland
T	Tristate

DEFECTFOUND

<u>DFTID</u>	ProductID	StoreID	DFTDate	DFTTime	NoOfItems
DFT101	1X1	S1	1-Jan-2020	8:00:00 AM	1
DFT202	2X2	S2	1-Jan-2020	8:30:00 AM	2
DFT303	3X3	S3	2-Jan-2020	8:45:00 AM	6

STORE

<u>StoreID</u>	StoreZip	RegionID
S1	60600	C
S2	60605	C
S3	35400	T

VENDOR

<u>VendorID</u>	VendorName
PG	Pacific Gear
MK	Mountain King

CATEGORY

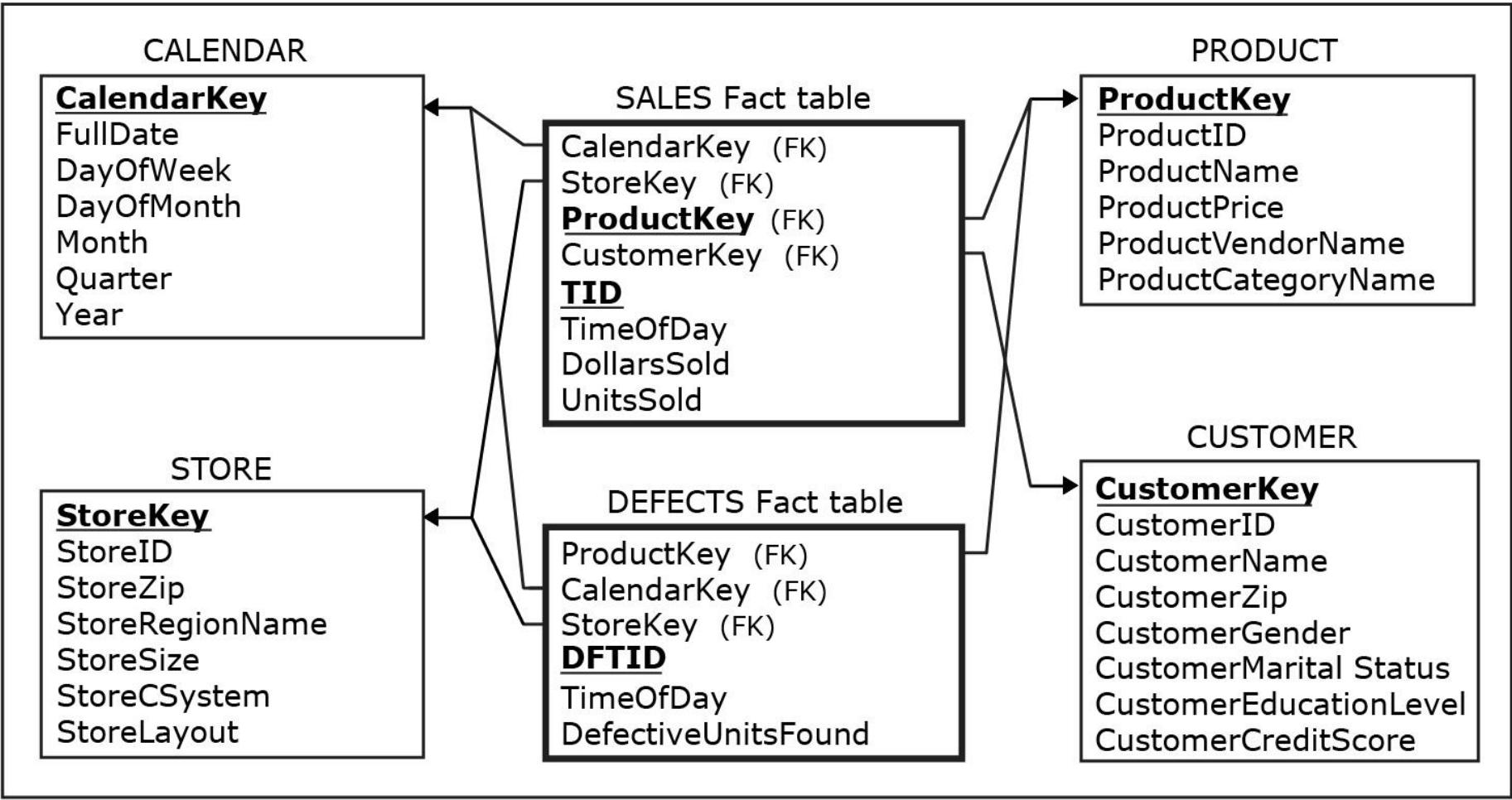
<u>CategoryID</u>	CategoryName
CP	Camping
FW	Footwear

PRODUCT

<u>ProductID</u>	ProductName	ProductPrice	VendorID	CategoryID
1X1	Zzz Bag	\$100	PG	CP
2X2	Easy Boot	\$70	MK	FW
3X3	Cosy Sock	\$15	MK	FW
4X4	Dura Boot	\$90	PG	FW
5X5	Tiny Tent	\$150	MK	CP
6X6	Biggy Tent	\$250	MK	CP

Expanded Example: Dimensional Model Based on Multiple Sources

ZAGI Retail Company dimensional model for the subjects *sales* and *defects*



Expanded Example: Dimensional Model Based on Multiple Sources

ZAGI Retail Company dimensional model for the subjects **sales** and **defects** , populated with the data from the four sources

CALENDAR Dimension

<u>Calendar Key</u>	FullDate	DayOf Week	DayOf Month	Month	Qtr	Year
1	1-Jan-2020	Wednesday	1	January	Q1	2020
2	2-Jan-2020	Thursday	2	January	Q1	2020

STORE Dimension

<u>Store Key</u>	StoreID	StoreZip	StoreRegion Name	StoreSize (m²)	Store CSystem	Store Layout
1	S1	60600	Chicagoland	51000	Cahiers	Modern
2	S2	60605	Chicagoland	35000	Self Service	Traditional
3	S3	35400	Tristate	55000	Mixed	Traditional

CUSTOMER Dimension

<u>Customer Key</u>	<u>CustomerID</u>	Customer Name	Customer Zip	Customer Marital Status	Customer Gender	Customer Education Level	Customer Credit Score
1	1-2-333	Tina	60137	Single	Female	College	700
2	2-3-444	Tony	60611	Single	Male	High School	650
3	3-4-555	Pam	35401	Married	Female	College	623

PRODUCT Dimension

<u>Product Key</u>	<u>ProductID</u>	Product Name	Product Price	ProductVendor Name	ProductCategory Name
1	1X1	Zzz Bag	\$100	Pacific Gear	Camping
2	2X2	Easy Boot	\$70	Mountain King	Footwear
3	3X3	Cosy Sock	\$15	Mountain King	Footwear
4	4X4	Dura Boot	\$90	Pacific Gear	Footwear
5	5X5	Tiny Tent	\$150	Mountain King	Camping
6	6X6	Biggy Tent	\$250	Mountain King	Camping

SALES Fact table

<u>Calendar Key</u>	<u>Store Key</u>	<u>Product Key</u>	<u>Customer Key</u>	<u>TID</u>	TimeOf Day	Dollars Sold	Units Sold
1	1	1	1	T111	8:23:59 AM	\$100	1
1	2	2	2	T222	8:24:30 AM	\$70	1
2	3	3	1	T333	8:15:08 AM	\$75	5
2	3	1	1	T333	8:15:08 AM	\$100	1
2	3	4	3	T444	8:20:33 AM	\$90	1
2	3	2	3	T444	8:20:33 AM	\$140	2
2	3	4	2	T555	8:30:00 AM	\$360	4
2	3	5	2	T555	8:30:00 AM	\$300	2
2	3	6	2	T555	8:30:00 AM	\$250	1

DEFECTS Fact table

<u>Calendar Key</u>	<u>Store Key</u>	<u>Product Key</u>	<u>DFTID</u>	TimeOfDay	DefectiveUnits Found
1	1	1	DFT101	8:00:00 AM	1
1	2	2	DFT202	8:30:00 AM	2
2	3	3	DFT303	8:45:00 AM	6

DIMENSIONAL MODELING

■ *Detailed versus aggregated fact tables*

- *Fact tables in a dimensional model can contain either detailed data or aggregated data*
- *In **detailed fact tables** each record refers to a single fact*
- *In **aggregated fact tables** each record summarizes multiple facts*

Detailed and Aggregated Fact Table Examples

ZAGI Retail Company
Sales Department
Database (Source 1)
with **additional** data
records included in
SALESTRANSACTION
and INCLUDES tables

REGION

<u>RegionID</u>	RegionName
C	Chicagoland
T	Tristate

STORE

<u>StoreID</u>	StoreZip	RegionID
S1	60600	C
S2	60605	C
S3	35400	T

PRODUCT

<u>ProductID</u>	Product Name	Product Price	VendorID	CategoryID
1X1	Zzz Bag	\$100	PG	CP
2X2	Easy Boot	\$70	MK	FW
3X3	Cosy Sock	\$15	MK	FW
4X4	Dura Boot	\$90	PG	FW
5X5	Tiny Tent	\$150	MK	CP
6X6	Biggy Tent	\$250	MK	CP

INCLUDES

<u>ProductID</u>	<u>TID</u>	Quantity
1X1	T111	1
2X2	T222	1
3X3	T333	5
1X1	T333	1
4X4	T444	1
2X2	T444	2
4X4	T555	4
5X5	T555	2
6X6	T555	1
5X5	T666	2
6X6	T666	1

SALESTRANSACTION

<u>TID</u>	CustomerID	StoreID	TDate	TTime
T111	1-2-333	S1	1-Jan-2020	8:23:59 AM
T222	2-3-444	S2	1-Jan-2020	8:24:30 AM
T333	1-2-333	S3	2-Jan-2020	8:15:08 AM
T444	4-5-666	S3	2-Jan-2020	8:20:33 AM
T555	2-3-444	S3	2-Jan-2020	8:30:00 AM
T666	2-3-444	S3	2-Jan-2020	9:30:00 AM

CATEGORY

<u>CategoryID</u>	Category Name
CP	Camping
FW	Footwear

VENDOR

<u>VendorID</u>	VendorName
PG	Pacifica Gear
MK	Mountain King

CUSTOMER

<u>CustomerID</u>	Customer Name	Customer Zip
1-2-333	Tina	60137
2-3-444	Tony	60611
3-4-555	Pam	35401

Detailed Fact Table Example

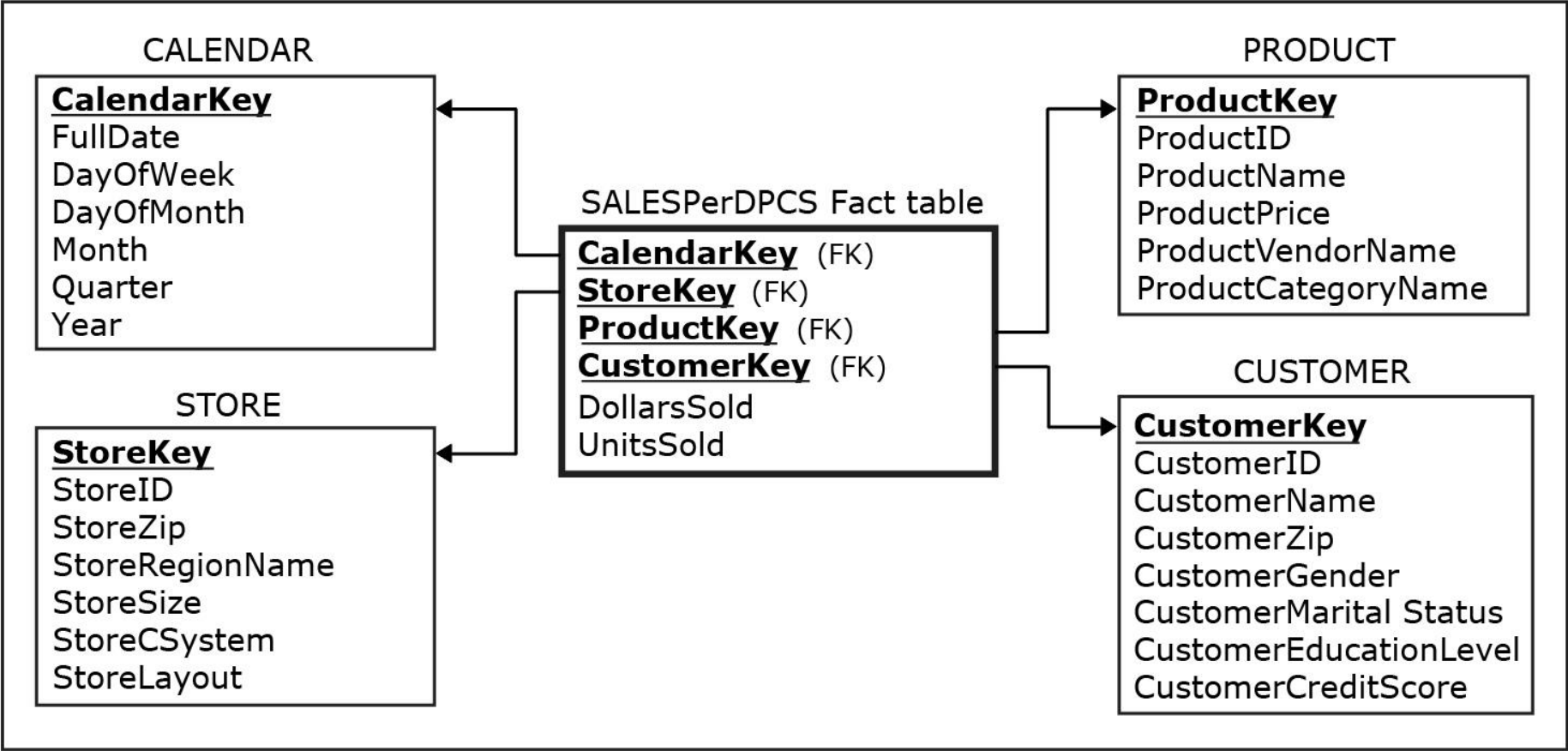
Detailed fact table in ZAGI Retail Company dimensional model for the subject **sales** with **additional** data included

SALES Fact table

Calendar Key	Store Key	Product Key	Customer Key	TID	TimeOfDay	Dollars Sold	Units Sold
1	1	1	1	T111	8:23:59 AM	\$100	1
1	2	2	2	T222	8:24:30 AM	\$70	1
2	3	3	1	T333	8:15:08 AM	\$75	5
2	3	1	1	T333	8:15:08 AM	\$100	1
2	3	4	3	T444	8:20:33 AM	\$90	1
2	3	2	3	T444	8:20:33 AM	\$140	2
2	3	4	2	T555	8:30:00 AM	\$360	4
2	3	5	2	T555	8:30:00 AM	\$300	2
2	3	6	2	T555	8:30:00 AM	\$250	1
2	3	5	2	T666	9:30:00 AM	\$300	2
2	3	6	2	T666	9:30:00 AM	\$250	1

Aggregated Fact Table Example 1

ZAGI Retail Company dimensional model with an aggregated fact table *Sales per day, product, customer, and store*



Aggregated Fact Table - Example 1

ZAGI Retail Company aggregated fact table for the subject sales:
Sales per day, product, customer, store, populated with the data

SALESPerDCS Fact table

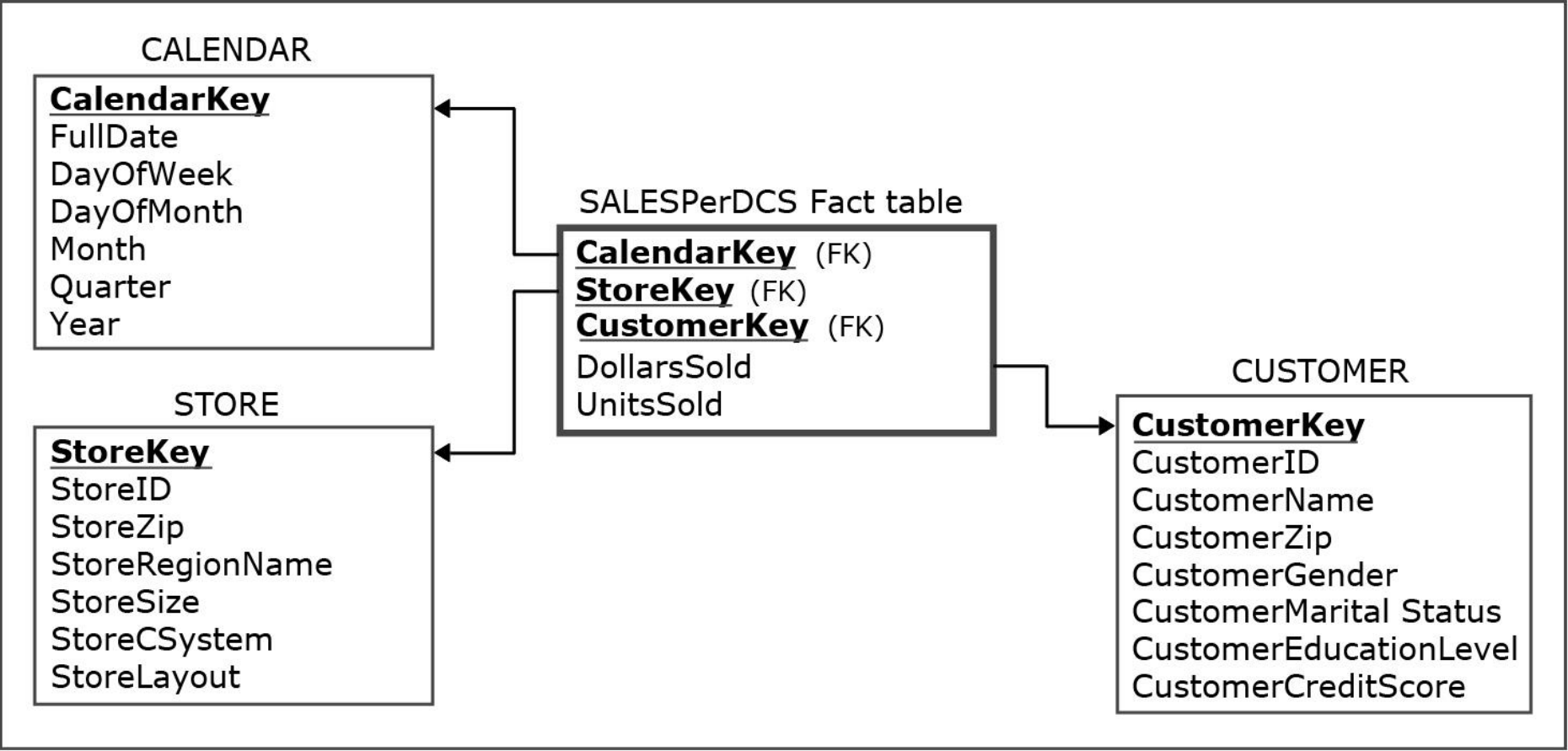
<u>Calendar Key</u>	<u>Store Key</u>	<u>Product Key</u>	<u>Customer Key</u>	Dollars Sold	Units Sold
1	1	1	1	\$100	1
1	2	2	2	\$70	1
2	3	3	1	\$75	5
2	3	1	1	\$100	1
2	3	4	3	\$90	1
2	3	2	3	\$140	2
2	3	4	2	\$360	4
2	3	5	2	\$600	4
2	3	6	2	\$500	2

← Amounts from 8th and 10th record in SALES fact table in Figure 8.22 combined (added)

← Amounts from 9th and 11th record in SALES fact table in Figure 8.22 combined (added)

Aggregated Fact Table - Example 2

ZAGI Retail Company star schema with an aggregated fact table *Sales per day, customer, and store*



Aggregated Fact Table - Example 2

ZAGI Retail Company aggregated fact table for the subject sales:
Sales per day, customer, store, populated with the data

SALESPerDCS Fact table

<u>Calendar Key</u>	<u>Store Key</u>	<u>Customer Key</u>	Dollars Sold	Units Sold
1	1	1	\$100	1
1	2	2	\$70	1
2	3	1	\$175	6
2	3	3	\$230	3
2	3	2	\$1,460	10

← Amounts from 3rd and 4th record in SALES fact table in Figure 8.22 combined (added)

← Amounts from 5th and 6th record in SALES fact table in Figure 8.22 combined (added)

← Amounts from 7th through 11th record in SALES fact table in Figure 8.22 combined (added)

DIMENSIONAL MODELING

■ *Granularity of the fact tables*

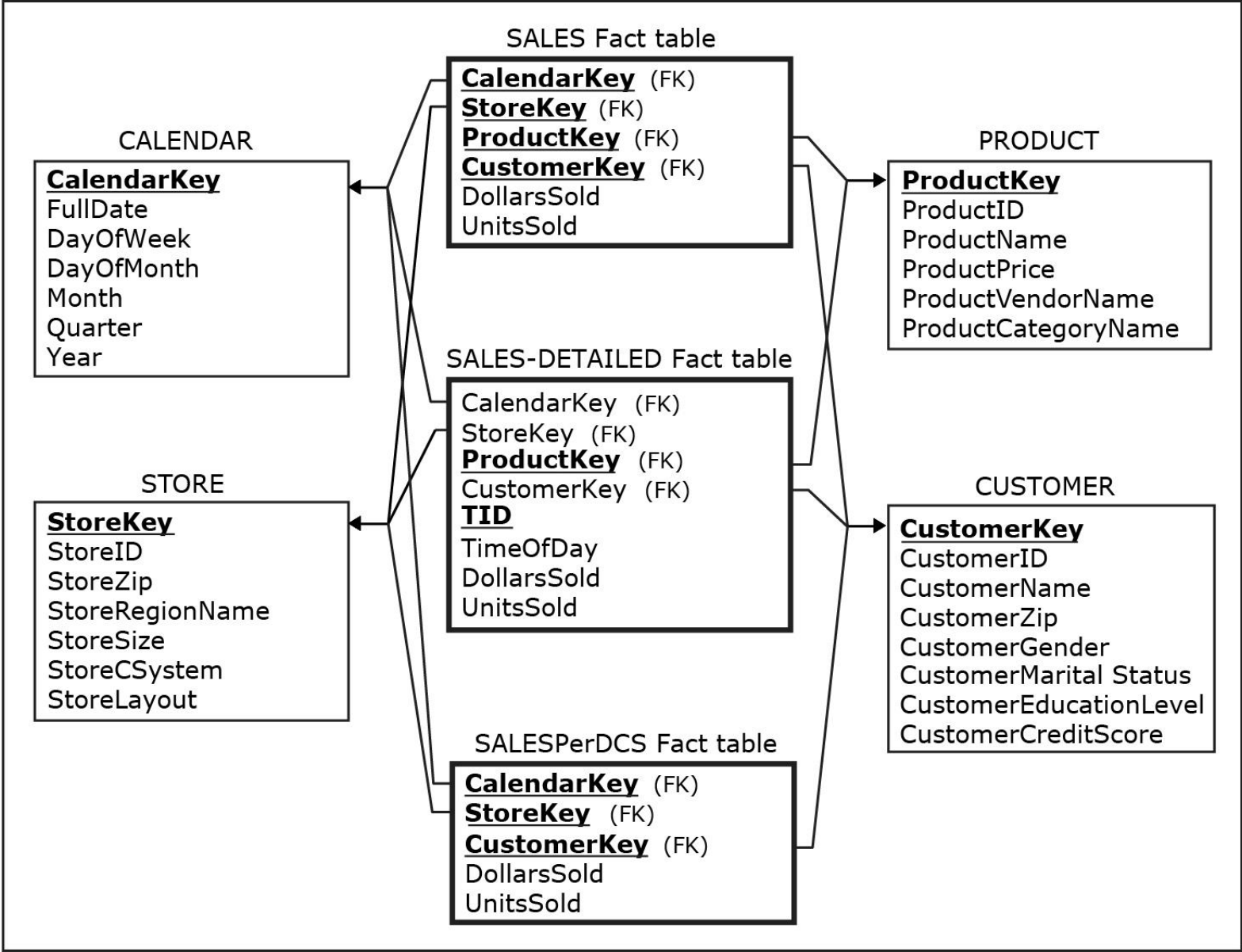
- *Granularity describes what is depicted by one row in the fact table*
- *Detailed fact tables have fine level of granularity because each record represents a single fact*
- *Aggregated fact tables have a coarser level of granularity than detailed fact tables as records in aggregated fact tables always represent summarizations of multiple facts*

DIMENSIONAL MODELING

■ *Granularity of the fact tables*

- *Due to their compactness, coarser granularity aggregated fact tables are quicker to query than detailed fact tables*
- *Coarser granularity tables are limited in terms of what information can be retrieved from them*
- *One way to take advantage of the query performance improvement provided by aggregated fact tables, while retaining the power of analysis of detailed fact tables, is to have both types of tables coexisting within the same dimensional model, i.e. in the same constellation*

A constellation of detailed and aggregated facts - Example



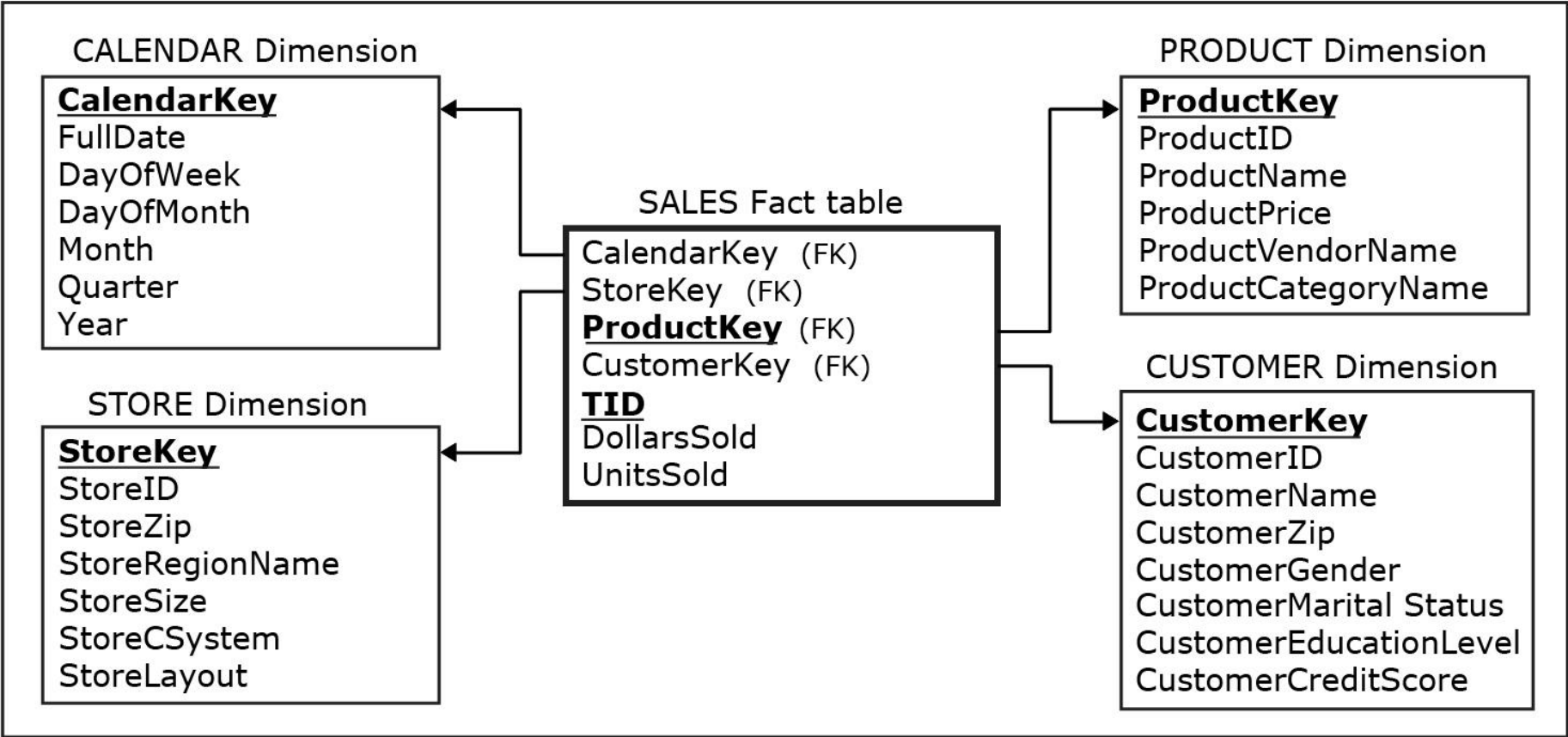
DIMENSIONAL MODELING

- *Tipos de detalhamento: Line-item versus transaction-level detailed fact table*
 - *Line-item detailed fact table*
 - Each row represents a line item of a particular transaction
 - *Transaction-level detailed fact table*
 - Each row represents a particular transaction

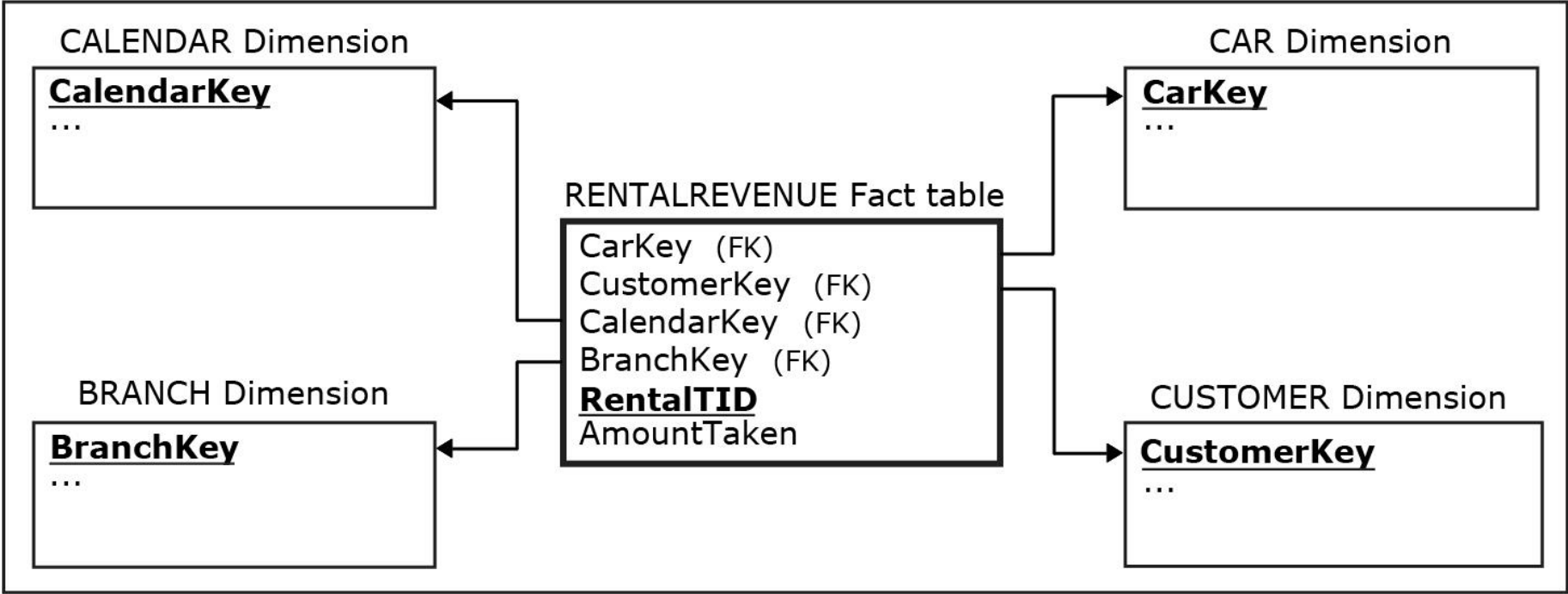
SALES Fact table

Calendar Key	Store Key	Product Key	Customer Key	TID	TimeOfDay	Dollars Sold	Units Sold
1	1	1	1	T111	8:23:59 AM	\$100	1
1	2	2	2	T222	8:24:30 AM	\$70	1
2	3	3	1	T333	8:15:08 AM	\$75	5
2	3	1	1	T333	8:15:08 AM	\$100	1
2	3	4	3	T444	8:20:33 AM	\$90	1
2	3	2	3	T444	8:20:33 AM	\$140	2
2	3	4	2	T555	8:30:00 AM	\$360	4
2	3	5	2	T555	8:30:00 AM	\$300	2
2	3	6	2	T555	8:30:00 AM	\$250	1
2	3	5	2	T666	9:30:00 AM	\$300	2
2	3	6	2	T666	9:30:00 AM	\$250	1

Line-Item Detailed Fact Table Example



Transaction-Level Detailed Fact Table Example



DIMENSIONAL MODELING

▪ *Slowly Changing Dimension*

- *Typical dimension in a star schema contains:*
 - **Attributes whose values do not change (or change extremely rarely) such as store size and customer gender**
 - **Attributes whose values change occasionally and sporadically over time, such as customer zip and employee salary.**
- *Dimension that contains attributes whose values can change referred to as a **slowly changing dimension***
- *Most common approaches to dealing with slowly changing dimensions*
 - **Type 1**
 - **Type 2**
 - **Type 3**

DIMENSIONAL MODELING

▪ *Type 1*

- *Changes the value in the dimension's record*
 - **The new value replaces the old value.**
- *No history is preserved*
- *The simplest approach, used most often when a change in a dimension is the result of an error*

Type 1 Example

Susan's Tax Bracket attribute value changes from *Medium* to *High*

CUSTOMER

<u>CustomerKey</u>	CustomerID	CustomerName	TaxBracket
1	111	Linda	Low
2	222	Susan	Medium
3	333	William	High

CUSTOMER

<u>CustomerKey</u>	CustomerID	CustomerName	TaxBracket
1	111	Linda	Low
2	222	Susan	High
3	333	William	High

DIMENSIONAL MODELING

▪ *Type 2*

- *Creates a new additional dimension record using a new value for the surrogate key every time a value in a dimension record changes*
- *Used in cases where history should be preserved*
- *Can be combined with the use of timestamps and row indicators*
 - **Timestamps - columns that indicates the time interval for which the values in the records are applicable**
 - **Row indicator - column that provides a quick indicator of whether the record is currently valid**

Type 2 Example

Susan's Tax Bracket attribute value changes from *Medium* to *High*

CUSTOMER

<u>CustomerKey</u>	CustomerID	CustomerName	TaxBracket
1	111	Linda	Low
2	222	Susan	Medium
3	333	William	High

CUSTOMER

<u>CustomerKey</u>	CustomerID	CustomerName	TaxBracket
1	111	Linda	Low
2	222	Susan	Medium
3	333	William	High
4	222	Susan	High

Type 2 Example (with timestamps and row indicator)
Susan's Tax Bracket attribute value changes from *Medium* to *High*

CUSTOMER			
<u>CustomerKey</u>	CustomerID	CustomerName	TaxBracket
1	111	Linda	Low
2	222	Susan	Medium
3	333	William	High

CUSTOMER						
<u>CustomerKey</u>	CustomerID	CustomerName	TaxBracket	Effective StartDate	Effective EndDate	Row Indicator
1	111	Linda	Low	1.1.2010	n/a	Current
2	222	Susan	Medium	1.1.2010	12.31.2019	Not current
3	333	William	High	1.1.2010	n/a	Current
4	222	Susan	High	1.1.2020	n/a	Current

DIMENSIONAL MODELING

▪ *Type 3*

- *Involves creating a “previous” and “current” column in the dimension table for each column where changes are anticipated*
- *Applicable in cases in which there is a fixed number of changes possible per column of a dimension, or in cases when only a limited history is recorded.*
- *Can be combined with the use of timestamps*

Type 3 Example

Susan's Tax Bracket attribute value changes from *Medium* to *High*

CUSTOMER			
<u>CustomerKey</u>	CustomerID	CustomerName	TaxBracket
1	111	Linda	Low
2	222	Susan	Medium
3	333	William	High

CUSTOMER				
<u>CustomerKey</u>	CustomerID	CustomerName	PreviousTax Bracket	CurrentTax Bracket
1	111	Linda	n/a	Low
2	222	Susan	Medium	High
3	333	William	n/a	High

Type 3 Example (with timestamps)
Susan's Tax Bracket attribute value changes from *Medium* to *High*

CUSTOMER

<u>CustomerKey</u>	CustomerID	CustomerName	TaxBracket
1	111	Linda	Low
2	222	Susan	Medium
3	333	William	High

CUSTOMER

<u>CustomerKey</u>	CustomerID	CustomerName	PreviousTax Bracket	Previous TaxBracket EffectiveDate	Current TaxBracket	Current TaxBracket EffectiveDate
1	111	Linda	n/a	n/a	Low	1.1.2010
2	222	Susan	Medium	1.1.2010	High	1.1.2020
3	333	William	n/a	n/a	High	1.1.2010

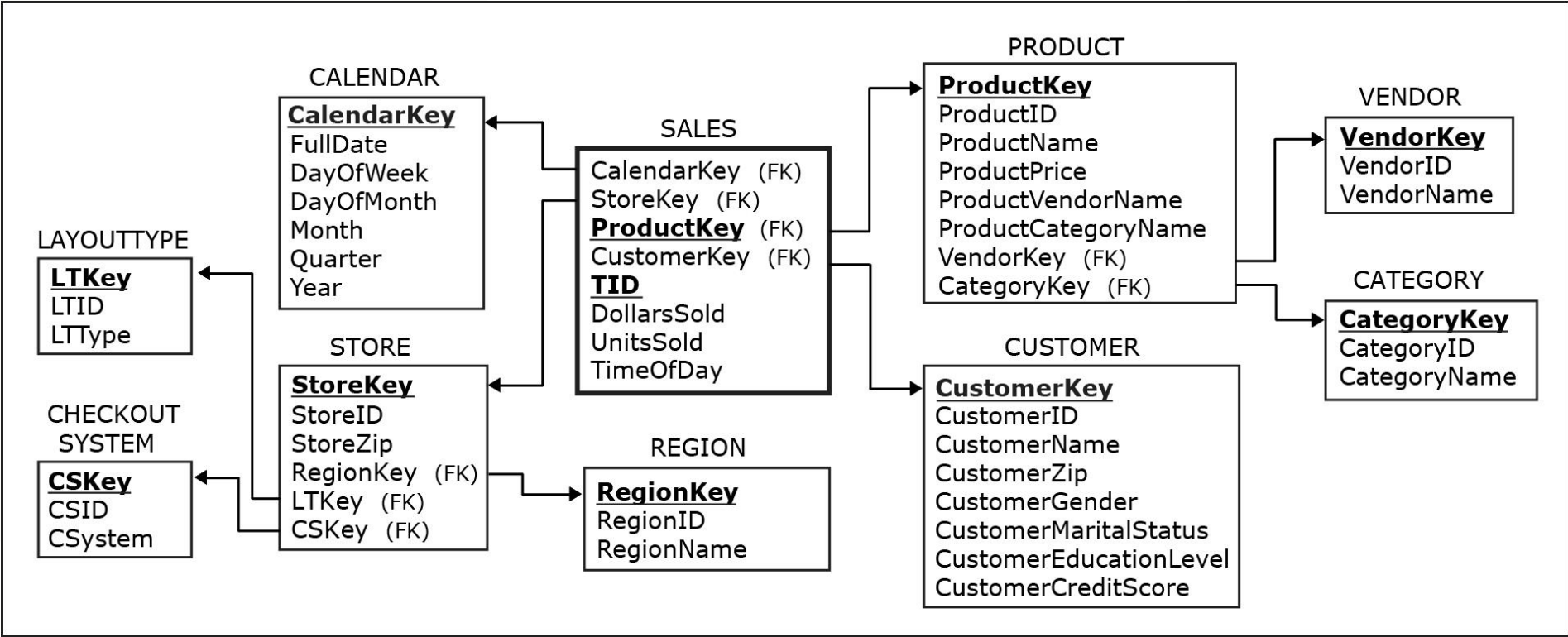
DIMENSIONAL MODELING

- **Snowflake model**

- *A star schema that contains the dimensions that are normalized*
- *Snowflaking is usually not used in dimensional modeling*
 - **Not-normalized (not snowflaked) dimensions provide for simpler analysis**
 - **Normalization is usually not necessary for analytical databases**

Snowflake Model - Example

A snowflaked version of the *ZAGI Retail Company* star schema for the subject sales



DATA WAREHOUSE (DATA MART) MODELING APPROACHES

- ***Data warehouse/data mart modeling approaches***
 - *Three of the most common data warehouse and data mart modeling approaches:*
 - **Normalized data warehouse**
 - **Dimensionally modeled data warehouse**
 - **Independent data marts**

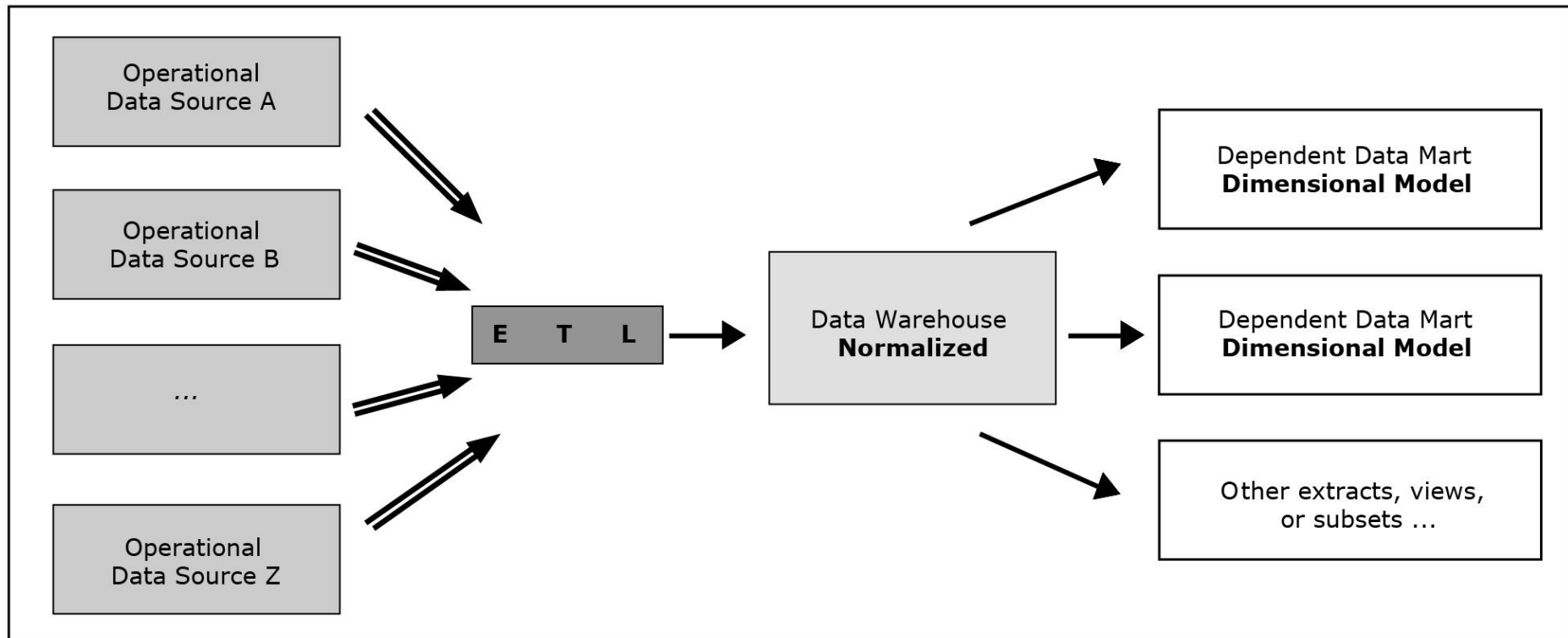
DATA WAREHOUSE (DATA MART) MODELING APPROACHES

■ *Normalized data warehouse*

- *Envisions a data warehouse as an integrated analytical database modeled by using the traditional database modeling techniques of ER modeling and relational modeling, resulting in a normalized relational database schema*
- *Populated with the analytically useful data from the operational data sources via the ETL process*
- *Serves as a source of data for dimensionally modeled data marts and for any other non-dimensional analytically useful data sets*

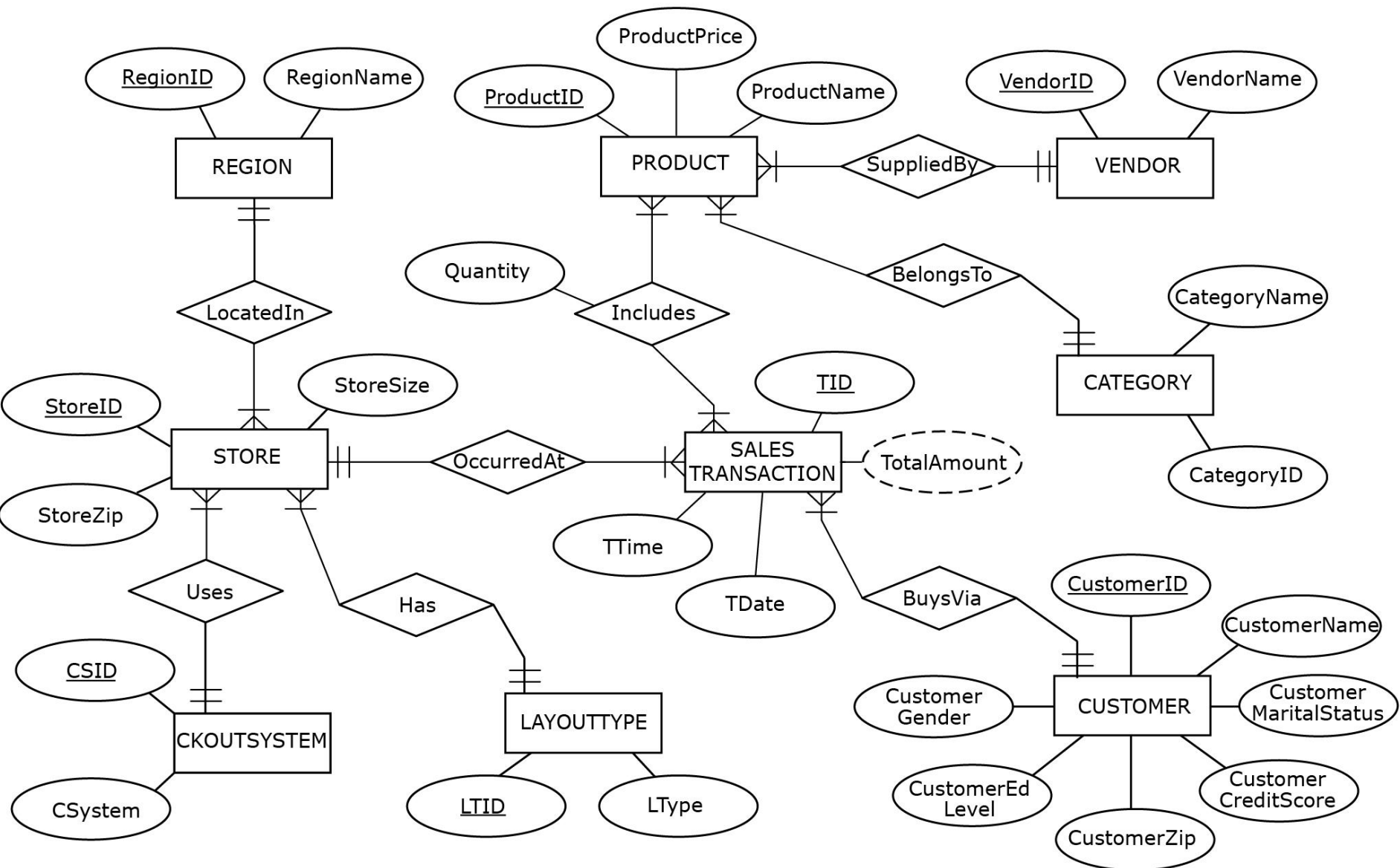
DATA WAREHOUSE (DATA MART) MODELING APPROACHES

- *Normalized data warehouse*



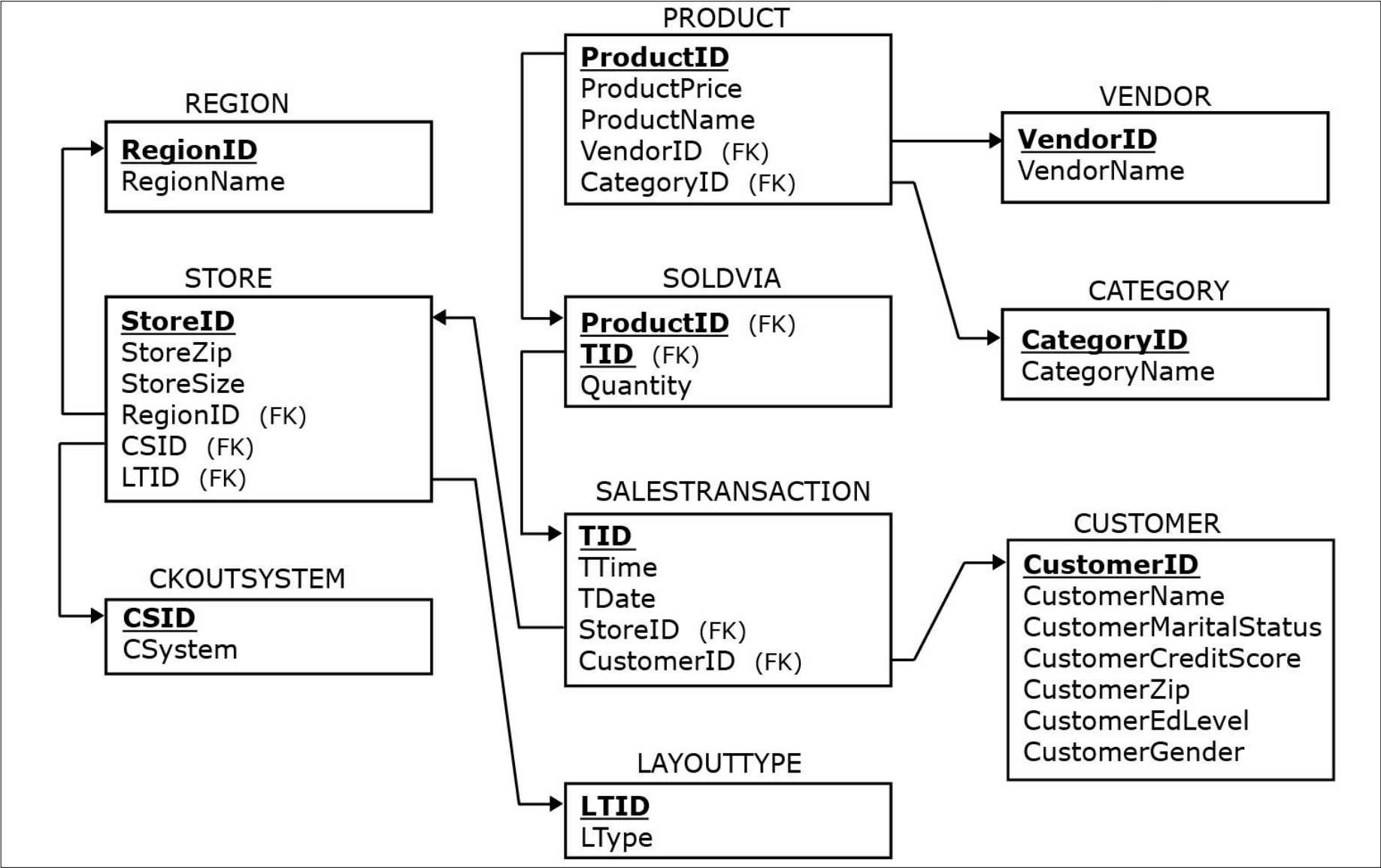
Normalized Data Warehouse - Example

ER-diagram: ZAGI Retail Company sales-analysis data warehouse



Normalized Data Warehouse - Example

Relational schema: *ZAGI Retail Company sales-analysis data warehouse*



Normalized Data Warehouse – Example

Data records:

ZAGI Retail Company sales-analysis data warehouse

PRODUCT

<u>ProductID</u>	Product Name	Product Price	VendorID	CategoryID
1X1	Zzz Bag	\$100	PG	CP
2X2	Easy Boot	\$70	MK	FW
3X3	Cosy Sock	\$15	MK	FW
4X4	Dura Boot	\$90	PG	FW
5X5	Tiny Tent	\$150	MK	CP
6X6	Biggy Tent	\$250	MK	CP

CATEGORY

<u>CategoryID</u>	Category Name
CP	Camping
FW	Footwear

CHECKOUTSYSTEM

<u>CSID</u>	CSSystem
CR	Cashiers
AC	Self Service
MX	Mixed

VENDOR

<u>VendorID</u>	VendorName
PG	Pacifica Gear
MK	Mountain King

REGION

<u>RegionID</u>	RegionName
C	Chicagoland
T	Tristate

LAYOUTTYPE

<u>LTID</u>	Layout
M	Modern
T	Traditional

STORE

<u>StoreID</u>	StoreZip	StoreSize (m ²)	RegionID	CSID	LTID
S1	60600	51000	C	CR	M
S2	60605	35000	C	AC	T
S3	35400	55000	T	MX	T

INCLUDES

<u>ProductID</u>	<u>TID</u>	Quantity
1X1	T111	1
2X2	T222	1
3X3	T333	5
1X1	T333	1
4X4	T444	1
2X2	T444	2
4X4	T555	4
5X5	T555	2
6X6	T555	1

SALESTRANSACTION

<u>TID</u>	CustomerID	StoreID	TDate	TTime
T111	1-2-333	S1	1-Jan-2020	8:23:59 AM
T222	2-3-444	S2	1-Jan-2020	8:24:30 AM
T333	1-2-333	S3	2-Jan-2020	8:15:08 AM
T444	3-4-555	S3	2-Jan-2020	8:20:33 AM
T555	2-3-444	S3	2-Jan-2020	8:30:00 AM

CUSTOMER

<u>CustomerID</u>	Customer Name	Customer Zip	Customer Gender	Customer Marital Status	Customer Education Level	Customer Credit Score
1-2-333	Tina	60137	Female	Single	College	700
2-3-444	Tony	60611	Male	Single	High School	650
3-4-555	Pam	35401	Female	Married	College	623

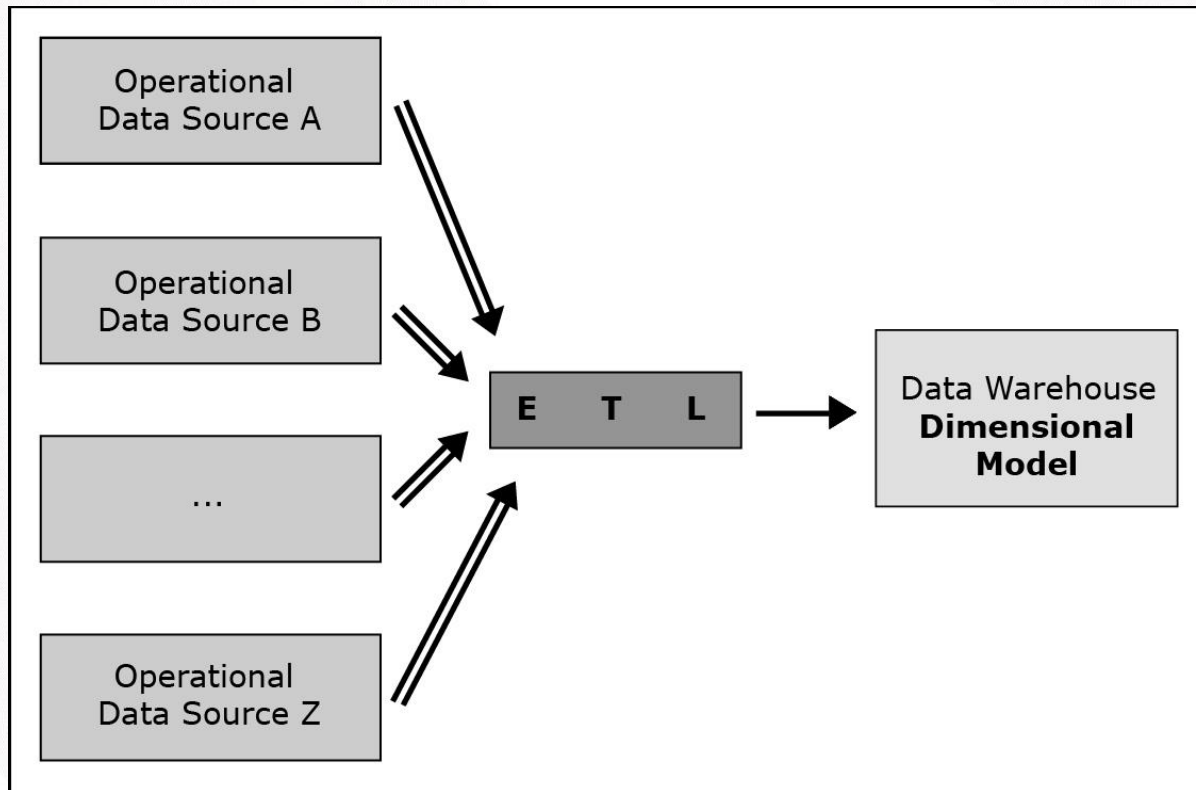
DATA WAREHOUSE (DATA MART) MODELING APPROACHES

- ***Dimensionally modeled data warehouse***

- *Data warehouse as a dimensional model that integrates analytically useful information from the operational data sources*
- *Same as the normalized data warehouse approach when it comes to the utilization of operational data sources and the ETL process*

DATA WAREHOUSE (DATA MART) MODELING APPROACHES

- *Dimensionally modeled data warehouse*



DATA WAREHOUSE (DATA MART) MODELING APPROACHES

▪ *Dimensionally modeled data warehouse*

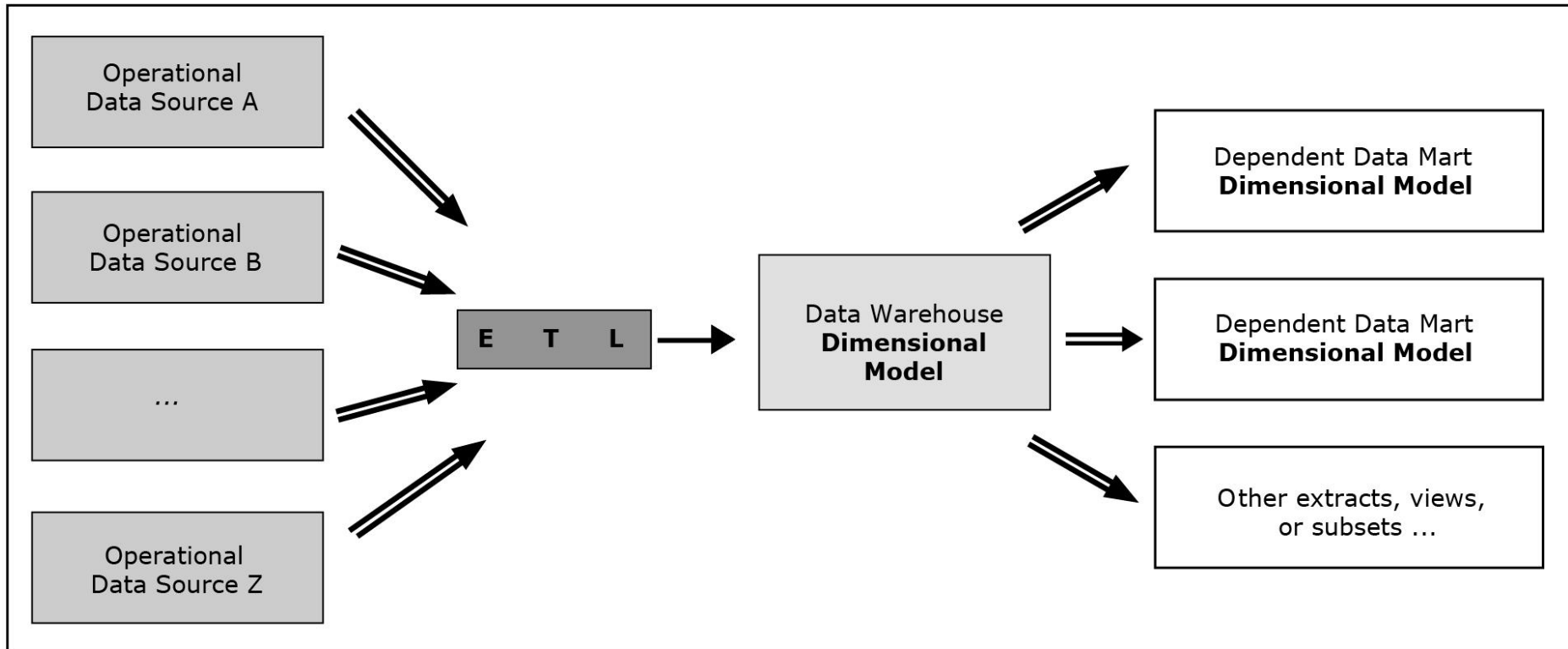
- *A set of commonly used dimensions known as **conformed dimensions** is designed first*
- *Fact tables corresponding to the subjects of analysis are then subsequently added*
- *A set of dimensional models is created where each fact table is connected to multiple dimensions, and some of the dimensions are shared by more than one fact table*
- *In addition to the originally created set of conformed dimensions, additional dimensions are included as needed*
- *The result is a data warehouse that is a dimensional model with multiple fact tables, i.e. a constellation of stars*

DATA WAREHOUSE (DATA MART) MODELING APPROACHES

- ***Dimensionally modeled data warehouse***
 - *Can be used as a source for dependent data marts and other views, subsets, and/or extracts*

DATA WAREHOUSE (DATA MART) MODELING APPROACHES

- *Dimensionally modeled data warehouse*



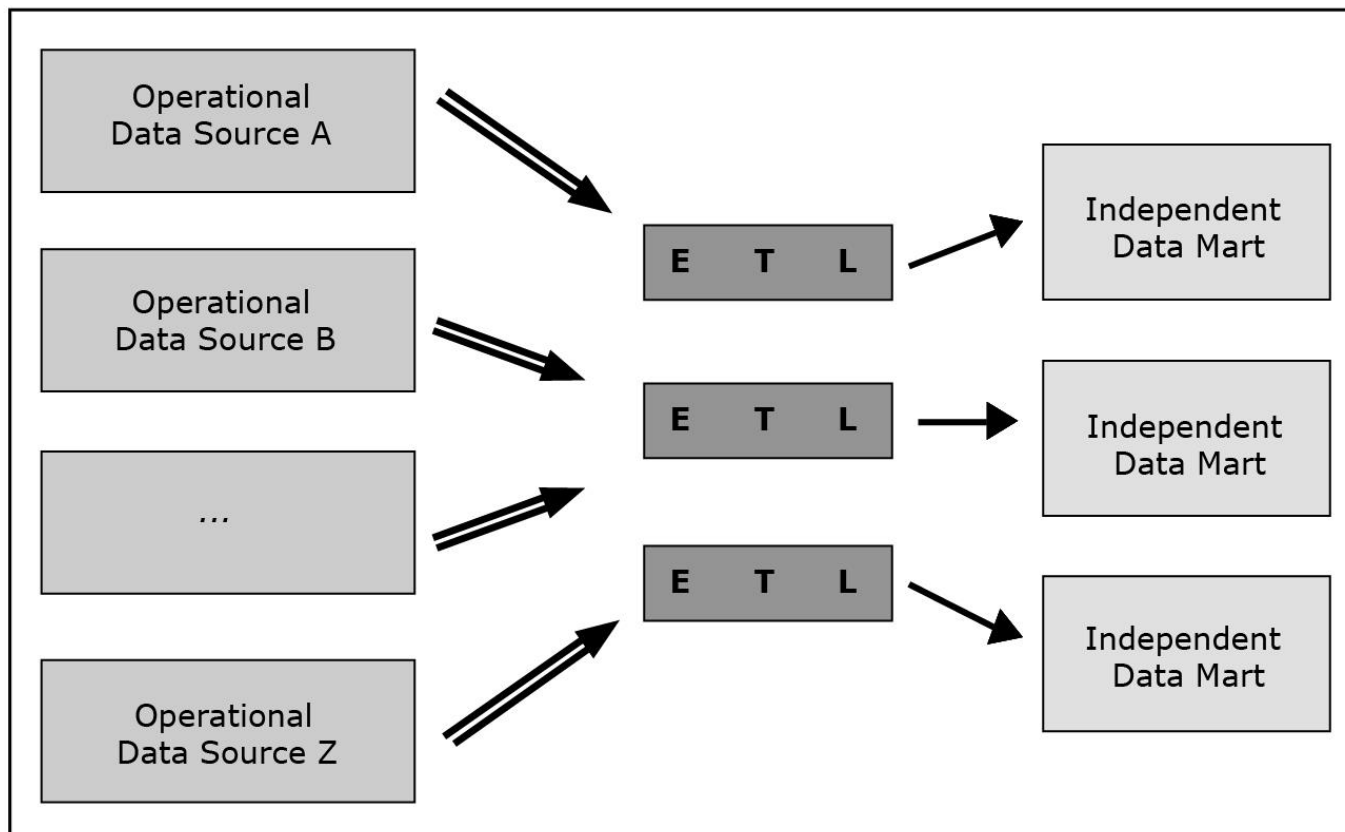
DATA WAREHOUSE (DATA MART) MODELING APPROACHES

- **Independent data marts**

- *Stand-alone data marts are created by various groups within the organization, independent of other stand-alone data marts in the organization*
- *Consequently, multiple ETL systems are created and maintained*

DATA WAREHOUSE (DATA MART) MODELING APPROACHES

- *Independent data marts*



DATA WAREHOUSE (DATA MART) MODELING APPROACHES

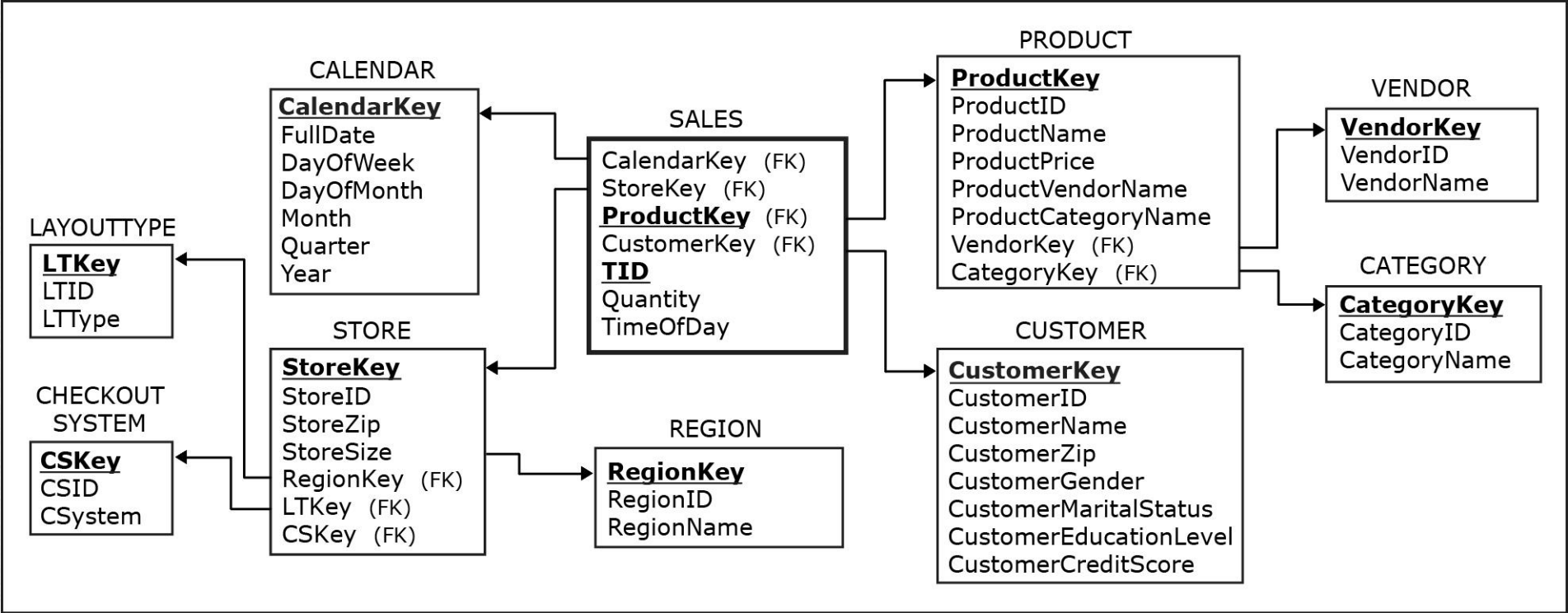
- ***Independent data marts***

- *Independent data marts are considered an inferior strategy*
 - **Inability for straightforward analysis across the enterprise**
 - **The existence of multiple unrelated ETL infrastructures**
- *In spite of obvious disadvantages, a significant number of corporate analytical data stores are developed as a collection of independent data marts*

DATA WAREHOUSE (DATA MART) MODELING APPROACHES

- *Comparing dimensional modeling and ER modeling as data warehouse/data mart design techniques*
 - *ER modeling can be used as a conceptual data warehouse/data mart design technique, followed by relational modeling as logical data warehouse/data mart design technique*
 - *Dimensional modeling can be used both for conceptual data warehouse/data mart design and logical data warehouse/data mart design*

Example - A modified normalized data warehouse schema



DATA WAREHOUSE (DATA MART) MODELING APPROACHES

- *Comparing dimensional modeling and ER modeling as data warehouse/data mart design techniques*
 - *Both ER modeling and dimensional modeling are viable alternatives for modeling data warehouses/data marts, and can be used within the same project*